Listening to History Rhymes: Pattern Recurrence and Return Predictability in Stock Markets

Abstract

Studies of price patterns have always had to say in advance which patterns to look for. We let the data say instead. We cut the daily return history of U.S. equities from 1970 to 2025 into roughly 1.32 million overlapping 60-day segments, standardize each one, and compare it against every other. A segment recurs when the real data are more crowded around it than a matched random-walk benchmark would imply, and the ratio of the two—the *Pattern Distance Ratio* (PDR)—turns recurrence into a continuous property of a return path that can be measured and tested. Recurrence is genuine and persistent: a segment's PDR in one year predicts its PDR in the next at a correlation of 0.308, still 0.270 three years out, and positive in every year of the sample, against 0.011 for a random-walk placebo. Recurrence also predicts returns. Segments that recur strongly carry more reliable directional forecasts, and a kernel ridge strategy built in trajectory space earns gross long–short alphas of roughly 6% to 9.5% per year against the CAPM and the Fama–French three- to six-factor models. The alphas hold in the most recent subperiods, and technical rules and standard predictors do not account for them. Deployed on twelve international markets with every design choice frozen at its U.S. value, the strategy earns a positive raw equal-weighted spread in all twelve, and a global country-average portfolio earns a significant alpha.

Keywords: return predictability, matrix profile, time series analysis, machine learning, technical analysis.

History never repeats itself, but it does often rhyme.

— Mark Twain

1 Introduction

Traditional finance theory, notably the efficient market and random walk hypotheses, has long provided a foundational framework for understanding stock price dynamics. While these theories offer valuable first-order approximations, a large body of literature has documented numerous market anomalies (e.g., Keim (1983), Ang et al. (2006), Banz (1981), Jegadeesh and Titman (1993)). Researchers have often relied on ex-ante hypotheses regarding human behavior (e.g., Stambaugh et al. (2012), Odean (1998), Shefrin and Statman (1985)), market imperfections (e.g., Duffie (2010)), or predefined statistical rules such as those used in technical analysis (e.g., Han et al. (2013), Zhu and Zhou (2009), Kwon and Kish (2002)) to uncover regularities in stock market dynamics. Nevertheless, our understanding of the intricate mechanisms driving markets remains limited, given the vast number of participants, the complexity of market mechanisms, and the multifaceted interactions among market components in determining prices.

More broadly, most empirical approaches in finance organize return data primarily through a panel-data perspective, where the primitive observational unit is a firm-time observation and statistical analysis proceeds through cross-sectional regressions, factor models, rolling-window statistics, or characteristic-based forecasting. While these approaches have generated important insights, the underlying representation of the data remains fundamentally tied to the stock-date panel structure.

The most direct antecedent of our approach is Lo et al. (2000), who put the reading of price patterns on a rigorous footing: they use nonparametric kernel smoothing to automate the identification of chart formations and then ask, through the conditional distribution of subsequent returns, whether those formations carry information. Their program established that visual pattern recognition can be made an algorithmic, statistically testable object rather than a matter of chartist judgment. It also left two questions open, both of which follow from the fact that the patterns themselves must still be named in advance. First, the search is confined to a dictionary of human-inherited shapes—head-and-shoulders, broadening tops, triangles—so a trajectory that recurs systematically but resembles no named formation is invisible to it, and one cannot ask which shapes recur, only whether the listed ones do. Second, because the object of study is a shape's average conditional return, the analysis stops short of the properties that determine whether such structure is economically meaningful: whether a segment's degree of recurrence is a stable characteristic that persists out of sample, and whether the structure survives the cost of trading on it. We take up the program from exactly this point. Replacing the predefined dictionary with an exhaustive comparison of every 60-day return trajectory against every other, we let the recurrent shapes emerge from the data; measuring each trajectory's recurrence against a matched random-walk benchmark, we obtain a continuous characteristic whose persistence can be tested directly; and carrying the analysis through to a traded portfolio charged at stock-level effective spreads, we can say not only whether recurrence predicts returns but whether it survives the frictions that any arbitrageur would face.

In this paper, we propose a complementary empirical perspective on stock return dynamics by reorganizing financial return data into a geometric space of standardized return trajectories. Rather than treating the stock-month observation as the primitive object of analysis, we partition the daily return history of U.S. equities into rolling 60-day return segments and embed these segments into a high-dimensional metric space, where similarity between trajectories is measured using z-normalized Euclidean distance. Under this representation, each observation becomes a return trajectory segment rather than a firm-date pair, and the geometry of the trajectory space itself becomes an object of empirical interest.

This representation opens a different class of empirical questions from those typically studied in traditional asset pricing and return predictability research. Instead of beginning with ex-ante hypotheses about particular economic mechanisms, firm characteristics, or predetermined chart patterns, our framework shifts attention toward the geometric organization of return trajectories themselves. This perspective naturally favors a more data-intensive and assumption-light approach, allowing large-scale structure to emerge from exhaustive comparison rather than from predefined conceptual classifications. It also enables the study of broader structural properties of the trajectory space that are difficult to observe within conventional panel-data representations.

Once return trajectories are embedded into a geometric similarity space, many additional questions become natural objects of study, including the topology of trajectory clusters, the density distribution of regions in trajectory space, the temporal evolution of neighborhoods, and the interaction between market regimes and geometric structure. The present paper focuses primarily on recurrence and predictability as initial probes into this space, leaving broader structural analysis for future research.

Motivated by these questions, we draw on the key insight of the *Matrix Profile* (Yeh et al., 2016), originally developed for motif discovery in time-series data (Lin et al., 2002): that exhaustive pairwise comparison of subsequences using z-normalized Euclidean distance can uncover meaningful recurring structures even in data with extremely low signal-to-noise ratios—a defining characteristic of financial returns. While the original Matrix Profile framework was designed primarily for motif discovery within a single time series, we extend its underlying principle to the full cross-section of U.S. equities, enabling systematic comparison across approximately 1.32 million rolling return segments spanning 1970–2025.

Importantly, our contribution is not merely the application of a new computational technique to an existing finance question. Rather, the Matrix Profile serves as the enabling methodology for constructing a new representation of financial return data. In this representation, recurrence, neighborhood structure, clustering, and trajectory geometry become natural empirical objects of study. Our analysis of pattern recurrence through the Pattern Distance Ratio (PDR) should therefore be viewed as an initial exploration of this broader geometric framework rather than as a complete characterization of the space.

Using nearest-neighbor distances from exhaustive pairwise comparisons, we construct the Pattern Distance Ratio (PDR), which quantifies pattern recurrence relative to a random walk benchmark constructed by reshuffling returns across stocks within each calendar year. The PDR compares how closely a given return segment is matched in the real data versus the randomized benchmark, imposing no ex-ante pattern classifications or predefined chart structures. The normalization by the random-walk benchmark is essential rather than cosmetic. Raw nearest-neighbor distances conflate genuine recurrence with the generic commonness of a segment's shape and with the changing density of the comparison set over time: smooth, trend-like trajectories are close to many segments in any data—real or random—and neighbor distances shrink mechanically as the cross-section grows. By measuring each segment's neighbor distances against a null constructed from the same returns, the same year, and the same search procedure, the PDR functions as a local density ratio—how much more crowded the real data are around a given trajectory than chance would imply—making recurrence comparable across shapes, stocks, and decades. Appendix D formalizes this interpretation in a latent-template model of recurring trajectories and derives the testable implications that organize our empirical analysis. This exhaustive, assumption-light search reveals that certain return trajectories recur with greater frequency and consistency than others. Throughout the paper, recurrence is therefore a statement about neighborhoods, not about exact repetition: no two 60-day trajectories are ever identical, and none need be. What recurs is the occupancy of regions in trajectory space—some neighborhoods attract far more segments than chance would imply, and, as we show below, the heavily visited neighborhoods remain heavily visited out of sample. History rhymes without repeating.

Our empirical analysis yields several findings. First, we document that pattern recurrence is genuine and persistent. Under annual comparison blocks, the correlation between in-sample and out-of-sample PDR is 0.308 at a one-year distance and decays only to 0.270 at a three-year distance, remaining positive in at least 94% of individual block-pairs at every lag—the slow decay of a structural characteristic of return dynamics rather than a short-lived phenomenon. Lengthening the comparison blocks to two, three, and five years raises the correlation to 0.486 and then 0.603, the precision gain from a larger comparison sample that our framework predicts.

A random-walk placebo, constructed by reshuffling returns across stocks within each year, produces a correlation of 0.011—roughly thirty times smaller, and never above 0.017 across the twenty placebo configurations our five independent shuffled series permit—ruling out the possibility that our results are artifacts of the distance metric, z-normalization procedure, or year-block comparison structure.

Transition matrices provide a complementary nonparametric perspective. Segments in the extreme PDR quintiles—both the most recurrent (Q1) and the most unique (Q5)—show the strongest tendency to remain in the same quintile over time, with Q5 retention probabilities ranging from 40.8% to 48.5% across horizons versus a null benchmark of 20%. The smooth off-diagonal decay indicates that movements within trajectory space are gradual rather than abrupt, consistent with PDR representing a slowly evolving structural characteristic of return trajectories. Importantly, the degree of persistence does not differ significantly between NBER recession and expansion periods, suggesting that the underlying mechanisms generating recurrent structures operate continuously across market environments.

We next establish a link between recurrence strength and return predictability. A sign-preservation analysis shows that strongly recurring segments maintain directional forecasts across time, with preservation rates of about 52–53% among strongly recurring segments with high neighbor-outcome scores—well above the independence-adjusted null of approximately 50.3%.

This systematic relationship between recurrence strength and predictability persistence suggests that the geometry of trajectory space captures economically meaningful regularities in stock return dynamics.

Finally, we translate these findings into a portfolio strategy. We map each return trajectory to its location in shape space—its Gaussian-kernel similarities to a grid of landmark trajectories drawn from the recurrence structure of Stage I—and predict future returns by ridge regression on this representation, in the spirit of the virtue-of-complexity literature (Kelly et al., 2024). Sorting firms each month into a long–short portfolio by predicted return, the strategy earns large gross risk-adjusted alpha—roughly 6% to 9.5% per year across the CAPM and Fama–French three- through six-factor models and equal- and value-weighted portfolios—that remains significant in the most recent subperiods. We then measure what it would cost to trade this signal, charging every trade at stock-level effective spreads estimated from daily high–low prices (Corwin and Schultz, 2012; Abdi and Ranaldo, 2017). The strategy's high turnover leaves it a break-even one-way trading cost of about 15 to 21 basis points; at our estimates of the spreads it faced the net-of-cost alpha is negative in every window, and the rolling gross return tracks the rolling cost of arbitrage over five decades. Rather than undermining the findings, this completes them—the recurrence-based mispricing persists precisely because it lives inside the no-arbitrage band that transaction costs create (Grossman and Stiglitz, 1980; Fama, 1991). Finally, we transplant the engine, with every design choice frozen at its U.S. value, to twelve international equity markets: the raw equal-weighted long–short return is positive in all twelve, a global country-average portfolio earns a significant alpha that survives restricting each market to its largest firms.

Technical analysis has consistently remained a preferred approach among investment practitioners. Menkhoff (2010) document the widespread reliance on technical analysis among fund managers across countries. On the academic side, while early research cast doubt on the statistical reliability of chartist approaches (e.g., Cowles (1933), Cowles (1944), Fama and Blume (1966)), subsequent studies have documented profitable trading rules and predictive structures (e.g., Brock et al. (1992), Bessembinder and Chan (1995), Lo et al. (2000)).

Our study aligns with this broad tradition in emphasizing time-series patterns in stock price movements. However, unlike conventional technical analysis, which typically relies on predetermined rules or handcrafted pattern definitions (e.g., Sullivan et al. (1999), Bajgrowicz and Scaillet (2012), Brock et al. (1992), Neely et al. (2014), Goulding et al. (2024)), our framework performs an exhaustive and assumption-light search over the space of return trajectories itself. In this sense, our methodology is fundamentally data-driven rather than rule-driven.

Our conclusion about tradability, by contrast, agrees with the most careful negative verdict in this literature. Bajgrowicz and Scaillet (2012) revisit the historical success of technical trading rules on the Dow Jones index, control for data snooping through the false discovery rate, show that an investor could not have selected the future best-performing rules ex ante, and find that the remaining performance is erased by low transaction costs. We reach the same verdict on tradability from a very different starting point—a cross-sectional search over 1.32 million individual return trajectories rather than a rule menu applied to one index, and stock-level effective spreads rather than a uniform cost assumption—and we regard the agreement as informative rather than redundant. The contribution here is not a tradable rule that survives where earlier ones failed; it is the object underneath. Whether a segment's shape recurs turns out to be a measurable, continuous, and strongly persistent property of the data—one that survives a block-bootstrap null, a diversified-neighbor null, and a placebo that severs shape from outcome, and that reappears in twelve international markets with every design choice frozen. That this property is priced in gross returns and then almost exactly offset by the cost of trading on it is a statement about how much structure the frictions of the market can leave standing, and it is the reason the two literatures converge.

In recent years, advances in machine learning and computational methods have renewed academic interest in nonlinear structure extraction from financial data. For example, Gu et al. (2020) demonstrate the predictive power of tree-based and neural-network methods relative to traditional linear models. Leippold et al. (2022) document substantial predictability in Chinese equity markets using machine-learning techniques. Jiang et al. (2023) employ convolutional neural networks to extract predictive information from transformed price-chart images, while Chen et al. (2024) develop adversarial deep-learning methods for conditional asset pricing.

Our study differs from this literature in an important respect. Much of the existing machine-learning literature improves predictive performance within conventional panel-data representations of financial data. By contrast, our contribution lies primarily at the level of data representation itself. Rather than transforming returns into image form or relying on firm characteristics as inputs, we organize financial return data directly into a geometric trajectory space and investigate the structural properties of that space.

Closest to our work is the contemporaneous study of Chen et al. (2026), who construct a high-dimensional economic-state vector from two centuries of newspaper text and show that the average market return following the most similar historical months forecasts the aggregate market—an explicit "history rhymes" design. Our paper shares the thesis but differs in every layer of implementation: we work in the cross-section of individual stocks rather than in aggregate market timing; our state object is the price trajectory itself rather than a news-derived description of the economic environment; and, before any prediction is attempted, our Stage I establishes that trajectory recurrence is a genuine, persistent, and placebo-robust property of the data—structure the prediction stage then exploits. The two papers also take opposite positions in an active methodological debate: Chen et al. (2026) deliberately keep the predictor simple—an equal-weighted average over nearest historical months—citing the caution of Nagel (2025) that high-dimensional kernel forecasts trained on short windows can degenerate into recency-weighted momentum, whereas our estimator retains the full high-dimensional similarity representation in the spirit of Kelly et al. (2024); Section 5.1 explains why the degeneracy identified by Nagel (2025) does not arise in our design. That two independent designs—one reading news text, the other reading price trajectories—converge on the same conclusion strengthens the case that recurring economic states leave predictable traces in returns.

While our approach retains the core principle of the Matrix Profile—exhaustive comparison of standardized subsequences via z-normalized Euclidean distance—our cross-sectional setting (approximately 1.32 million segments across 55 years) necessitates a departure from the standard FFT-based implementation. We instead compute distances directly using GPU-accelerated matrix operations, which accommodate our large-scale cross-sectional comparison structure while maintaining computational feasibility.

Our study makes several contributions to the asset pricing literature. First, inspired by the Matrix Profile's principle of exhaustive subsequence comparison, we develop a scalable distance-based methodology for organizing and analyzing the geometric structure of stock return trajectories across the full cross-section of U.S. equities. Second, we introduce the Pattern Distance Ratio (PDR)—a self-normalized local density ratio that measures each segment's recurrence against a shape- and year-specific random walk null—and provide evidence that recurrence in trajectory space is genuine and persistent. Third, we establish a direct link between recurrence strength and the persistence of return predictability. Finally, we demonstrate that information embedded in the geometric structure of return trajectories generates significant abnormal returns that remain unexplained by standard factor models.

The remainder of the paper is organized as follows. Section 2 describes the data and the Matrix Profile-inspired methodology for constructing the trajectory space and measuring recurrence. Section 3 presents the central empirical results on recurrence and persistence, together with robustness checks. Section 4 links recurrence strength to the persistence of return predictability. Section 5 develops the trajectory-space portfolio strategy and examines its performance, the role of the recurrence structure, the economic interpretation of the alpha, its trading costs and capacity, and its one-shot deployment in international markets. Section 6 concludes.

2 Data and Matrix Profile-Inspired Measurement of Price Pattern Recurrence

Figure 1 provides a high-level overview of our methodological pipeline, from raw data processing to portfolio construction. The remainder of this section lays out the construction steps of the first stage of this framework—the identification of recurrent price patterns.

2.1 Data

Our analysis draws on three primary data sources: (i) daily stock market data from the Center for Research in Security Prices (CRSP), (ii) annual accounting data from Compustat, and (iii) daily and monthly Fama-French factor returns obtained from Kenneth French's data library via the Wharton Research Data Services (WRDS) platform. The sample covers the period from January 1970 to December 2025. Below we describe each step of the sample construction in detail.

We obtain daily stock returns, prices, and shares outstanding from CRSP for all common stocks (share codes 10 and 11) listed on the NYSE, AMEX, and Nasdaq (exchange codes 1, 2, and 3). Daily returns incorporate delisting returns following the standard procedure. Market equity is defined as $ME_{j,t} = |P_{j,t}| \times SHROUT_{j,t}$,[1] aggregated across all share classes within each firm (*permco*). Book equity is constructed from Compustat following Fama and French (1993) as $BE = SEQ + TXDITC - PS$, with the standard preferred stock hierarchy (redemption, liquidating, then par value). We require $BE > 0$ and at least two years of Compustat data to

[1] All variables and notations are defined in Table A1 in the Appendix.

mitigate backfill bias.

To construct characteristic-adjusted returns, we form 25 benchmark portfolios by independently sorting stocks into quintiles based on market capitalization (size) and the book-to-market equity ratio (BM), following the methodology of Fama and French (1993).

We apply two filters to mitigate the influence of illiquid or extremely small stocks, which may generate spurious patterns driven by microstructure noise rather than genuine price dynamics. We exclude all stock-year observations for which the stock's closing price at the end of June is below $5. This filter removes penny stocks whose returns may be dominated by bid-ask bounce and discrete price effects. We exclude all stock-year observations for which the stock's market equity at the end of June falls below the 10th percentile of NYSE-listed firms' market equity in the same month. This filter removes microcap stocks that may lack sufficient trading activity to generate reliable price patterns. Both filters are applied annually: a stock excluded in one July-to-June period may re-enter the sample in a subsequent period if it satisfies the criteria. After applying these filters, our final sample spans 55 years (1971–2025) and contains approximately 1.32 million return segments.

2.2 Return Segments and Matrix Profile Distance Metrics

Our primary objects of interest are individual stock daily return segments constructed using a rolling-window approach. To keep the sample size manageable, the windows roll at a monthly frequency and are anchored to *month-end dates*. Specifically, for each month-end trading day t in the sample, we define a *formation window* consisting of the preceding 60 trading days up to and including t. The price movement segments derived from this formation window serve as inputs to the Matrix Profile procedure in our main stage of analysis, where we identify recurring price dynamic patterns.

In the second stage, which we explore later in the paper, we examine the investment implications of these patterns within a return predictability framework. For this purpose, we define an *evaluation window* consisting of the subsequent 20 trading days immediately following the *formation window*.

Return Segments

Each return segment in the *formation window* is uniquely identified by the pair (i, t), where i denotes the stock and t the window-ending date. The corresponding 60-day return time series is

$$\mathbf{R}_{i,t} = \left(r_{i,t-59},\ r_{i,t-58},\ \ldots,\ r_{i,t}\right), \tag{1}$$

where $r_{i,s}$, for $s = t - 59, \ldots, t$, denotes the daily return of stock i on trading day s. We require each stock to have a complete set of 60 returns within the *formation window*; segments with missing returns are excluded. For brevity, we often index segments by a single index n, with $n = (i, t)$. When needed, the underlying stock or date is written as $i(n)$ and $t(n)$, or simply i and t.

The Matrix Profile Framework and Distance Metric

Our distance-based approach builds on the Matrix Profile, a time series analysis technique introduced by Yeh et al. (2016) that extends earlier work on motif discovery (Lin et al., 2002). The Matrix Profile computes the z-normalized Euclidean distance between all pairs of subsequences in a time series to identify recurring subsequences, or "motifs." For two subsequences $T_{i,m}$ and $T_{j,m}$ of length m, the z-normalized Euclidean distance is defined as

$$d_{i,j}^{MP} = \sqrt{2m\left(1 - \frac{QT_{i,j} - m\,\mu_i\,\mu_j}{m\,\sigma_i\,\sigma_j}\right)}, \tag{2}$$

where μ_i, μ_j and σ_i, σ_j are the means and standard deviations of the two subsequences, and $QT_{i,j}$ is their dot product. This distance is equivalent to the Euclidean distance between the two subsequences after each has been standardized to zero mean and unit variance. Subsequences with small z-normalized distances share similar temporal dynamics after controlling for differences in level and scale.

We compute the z-normalized Euclidean distance d^{MP}, as defined in Equation (2), for the formation window return segments. For 60-day segments indexed by $m = (i, t)$ and $n = (j, s)$, we denote this distance by $d_{m,n}$ for brevity.[2] A smaller value of $d_{m,n}$ indicates that the two segments exhibit similar temporal dynamics, after accounting for differences in mean return and

[2]Since both vectors are standardized, this distance is monotonically related to the Pearson correlation: $d_{m,n}^2 = 2 \times 60 \times (1 - \hat{\rho}_{m,n})$.

volatility.

The Matrix Profile algorithm relies on the Fast Fourier Transform (FFT) for efficient all-pairs comparisons within a single time series. Our setting, however, compares return segments *across* the entire cross-section of stocks—about 1.32 million observations—requiring hundreds of billions of pairwise distance evaluations to random walk benchmark series. This cross-sectional structure does not naturally map onto the single-series framework for which FFT-based Matrix Profile algorithms are optimized. To make the computation tractable, we implement GPU-accelerated matrix operations using the `CuPy` library.[3]

2.3 Random Walk Benchmark

To assess whether the pattern similarities observed in real stock returns reflect genuine recurrence or arise merely by chance, we construct a benchmark from randomly shuffled returns. The null this benchmark embodies should be stated precisely. Pooling and reshuffling all returns within a calendar year produces series that are temporally unstructured *and* cross-sectionally homogeneous: every synthetic stock-day is an i.i.d. draw from the year's pooled return distribution. This null is deliberately stronger than the random-walk hypothesis itself, which is silent on distributional homogeneity—a random walk is fully compatible with heteroskedasticity, stochastic volatility, jumps, and stock-level heterogeneity—so real data can in principle deviate from the pooled-i.i.d. benchmark through channels other than recurring trajectory shapes. Accordingly, the shuffle benchmark plays two distinct roles in this paper, and it is important to keep them separate. Its primary role is as the *normalization device* inside the PDR: it controls for a segment's generic shape commonness, the size of the comparison set, and the year's marginal return distribution (Section 2.4), for which the pooled construction is exactly what is needed. The role of *null hypothesis for recurrence*, by contrast, is ultimately delegated to a far more demanding benchmark—the stationary block-bootstrap surrogate of Section 3.3, which preserves each stock's own return distribution (including its skewness, tails, and jump frequency), its short-range dependence such as volatility clustering, and its factor comovement, and against which both the persistence results and the recurrence–predictability gradient survive. Under the shuffle benchmark, if returns carried no temporal or cross-sectional structure, the distances between standardized return segments

[3] All GPU computations are conducted on a system equipped with an AMD Ryzen 9 9950X3D processor and three NVIDIA GeForce GPUs (an RTX 5090, an RTX 5080, and an RTX 4090), across which the distance computations are distributed, running CUDA 12.8 and CuPy 13.6.0. Distance matrices are stored in single-precision floating-point (`float32`) format, retaining approximately 7 significant decimal digits of precision.

in the real data would be statistically indistinguishable from those computed against the shuffled series; systematic deviations indicate the presence of structure, and the surrogate then isolates how much of it is genuine shape recurrence rather than distributional heterogeneity or simple autocorrelation.

We construct the random-walk benchmark using a *cross-stock randomization* procedure. All daily returns from all stocks within each calendar year are pooled and randomly redistributed across all stock-date observations. Formally, let

$$\mathcal{R}_y = \left\{ r_{j,t} : \text{stock } j \text{ is in the sample on trading day } t \text{ in calendar year } y \right\} \tag{3}$$

denote the complete set of observed daily returns across all stocks in year y, and let $N_y = |\mathcal{R}_y|$ be the total number of returns that year. We draw a random permutation σ of $\{1, 2, \ldots, N_y\}$ and assign the $\sigma(k)$-th element of $\mathcal{R}_y$ to the k-th stock-date observation. In other words, each observation (j, t) in year y receives a return drawn uniformly at random, without replacement, from the pool of all returns observed across all stocks in that year.

We generate five independent random-walk series by repeating the shuffling procedure five times. The first three series ($m = 1, 2, 3$) serve as the benchmark for computing the PDR denominator (Section 2.4), while the fifth series ($m = 5$) plays the pretend-real query role in the placebo test; the remaining series allow the placebo to be re-formed under the twenty alternative query–benchmark configurations examined in Section 3.1.

2.4 Pattern Distance Ratio (PDR)

To quantify pattern recurrence, we compare the similarity of each real return segment to other real segments versus its similarity to segments from the bootstrap-generated random walk sample described above. The key idea is that if a pattern genuinely recurs in real data, its nearest neighbors among other return segments should be systematically closer (i.e., have smaller L2 distances) than its nearest neighbors in the random-walk sample.

To structure the pairwise comparisons and align with the persistence analysis in Section 3, we partition all return segments into annual data blocks. Each segment is assigned to the calendar year in which its *formation window* ends. Let $\mathcal{D}_y$ denote the set of all segments whose 60-day window concludes in year y, for $y = 1971, 1972, \ldots, 2025$.

For each annual block $\mathcal{D}_y$, we compute two types of pairwise distance matrices:

- **Within-block (in-sample):** L2 distances between every pair of segments (n, n') with $n, n' \in \mathcal{D}_y$, producing the distance matrix for $\mathcal{D}_y$ versus $\mathcal{D}_y$.
- **Across-block (out-of-sample):** L2 distances between each segment $n \in \mathcal{D}_y$ and every segment $n' \in \mathcal{D}_{y+h}$ for lags $h = 1, 2, 3$, producing the distance matrix for $\mathcal{D}_y$ versus $\mathcal{D}_{y+h}$.

Two versions of the across-block comparison appear in the persistence analysis, and it is important to distinguish them. In the *block-length* specifications (the 1-Year, 2-Year, 3-Year, and 5-Year columns of Table 1, Panels A–C), the sample is repartitioned into non-overlapping one-, two-, three-, or five-year blocks and each block is compared with the next; lengthening the block therefore increases the temporal distance between the compared blocks *and* the size of the comparison sample simultaneously. In the *fixed-block, varying-lag* specification (Panel D of the same table), the blocks are held at one year and only the lag h is varied, which isolates the effect of temporal distance from that of the comparison-sample size.

These pairwise distance computations are performed five times in parallel: once using real standardized returns for both segments (yielding the Real-to-Real distances) and once for each of the four random walk series described below (yielding Real-to-Random distances, where the query segment uses real returns and the candidate segment uses random returns).

Return segments from different stocks may exhibit comovements and low L2 distances due to common systematic factors. To mitigate this, we exclude from the pairwise calculations any two segments that share the same formation window. Since segments are constructed using a monthly rolling window, this exclusion ensures that all compared segments have at least a one-month gap between the starting dates of their respective formation windows.

For each real segment n, we identify its $K = 100$ nearest neighbors based on the z-normalized Euclidean distance (Equation 2). We then compute two types of mean nearest-neighbor distances for segment n:

1. **Real-to-Real Distance** ($\bar{d}_n^{real \to real}$): Among all other real return segments (excluding those on the same calendar date as n), we identify the K segments with the smallest L2 distances to n. The Real-to-Real Distance is the mean L2 distance to these nearest neighbors:
$$\bar{d}_n^{real \to real} = \frac{1}{K} \sum_{k=1}^{K} d_{n, \nu_k^{real}}, \tag{4}$$

where v_k^{real} is the k-th nearest neighbor of n among all other real segments. Smaller values of $\bar{d}_n^{real\to real}$ indicate closer matches in real data, suggesting frequent recurrence.

2. **Real-to-Random Distance** ($\bar{d}_n^{real\to rand}$): For each of the three random walk series $m \in \{1, 2, 3\}$, we compute the L2 distances between the real segment n and all segments from random series m, identify the K nearest neighbors, and compute the mean distance $\bar{d}_n^{real\to rand,m}$. The Real-to-Random Distance is the average across the three series:

$$\bar{d}_n^{real\to rand} = \frac{1}{3}\sum_{m=1}^{3} \bar{d}_n^{real\to rand,m}, \quad \text{where} \quad \bar{d}_n^{real\to rand,m} = \frac{1}{K}\sum_{k=1}^{K} d_{n,\, v_k^{rand,m}}. \tag{5}$$

   Here, $v_k^{rand,m}$ denotes the k-th nearest neighbor of n among segments from random series m.

The **Pattern Distance Ratio (PDR)** for segment n is defined as

$$PDR_n = \frac{\bar{d}_n^{real\to real}}{\bar{d}_n^{real\to rand}}, \tag{6}$$

where $\bar{d}_n^{real\to real}$ (Equation 4) and $\bar{d}_n^{real\to rand}$ (Equation 5) are the mean distances from n to its K nearest neighbors in real and random data, respectively. PDR compares how closely a segment is matched in the actual market versus a random-walk benchmark.

If $PDR_n < 1$, the nearest neighbors of segment n in real data are closer than those in random data, indicating genuine pattern recurrence. Conversely, if $PDR_n > 1$, segment n is less similar to other real segments than to random segments, suggesting relative uniqueness in the real data.

The two ends of the PDR distribution make this construction concrete. For the most recurrent segment of an annual block, the real neighbors trace visibly tighter paths around the query than the random-walk neighbors do; for the most novel segment—an idiosyncratic jump-like trajectory—the ordering reverses, and shuffled data matches the shape more closely than real data does. The PDR summarizes exactly this contrast in a single number per segment.

Why Normalize by the Random-Walk Benchmark?

The denominator of the PDR is central to the construction, not a cosmetic adjustment. The raw nearest-neighbor distance $\bar{d}_n^{real\to real}$ is not, by itself, a valid measure of recurrence, because the expected distance from a segment to its nearest neighbors varies mechanically with factors unrelated to genuine pattern repetition. First, even after z-normalization, different trajectory shapes occupy regions of the shape space with very different null densities: smooth, trend-like segments are generically close to many other segments—real or random alike—whereas jagged, high-frequency segments are generically far from every comparison segment. A ranking based on the numerator alone would therefore in large part sort segments by this generic shape commonness rather than by recurrence. Second, the density of the comparison set varies over time: the cross-section of U.S. equities—and hence the number of candidate neighbors in each annual block—changes substantially over the 55-year sample, and nearest-neighbor distances shrink mechanically as the comparison set grows. Third, the marginal distribution of daily returns itself differs across years and market environments, shifting the entire distance distribution.

The Real-to-Random distance in the denominator absorbs all three effects by construction. Because the random-walk segments are built from the same pool of returns within the same calendar year, standardized by the same within-window procedure, and searched with the same K nearest neighbors over a comparison set of the same size, $\bar{d}_n^{real\to rand}$ measures how close segment n would be to its neighbors under the null of no temporal structure—conditional on its own shape, the size of the comparison set, and the year's return distribution. The ratio therefore admits a natural interpretation as a *local density ratio*: since nearest-neighbor distances are inversely related to local density (Loftsgaarden and Quesenberry, 1965), PDR compares the density of real trajectories in the neighborhood of segment n with the density of null trajectories at the same location in shape space. $PDR_n < 1$ says that the region of trajectory space around n is more densely populated in real data than chance would imply—genuine recurrence—rather than merely that n has a common shape or sits in a well-sampled year. It is this self-normalization that makes PDR comparable across segments, stocks, years, and volatility regimes, and thereby allows us to pool approximately 1.32 million segments spanning 1971–2025 into a single analysis.

Appendix D formalizes this interpretation in a latent-template mixture model of return trajectories. In the model, the probability limit of the PDR is an exact monotone transform of the local density ratio; the raw neighbor distance is confounded by generic shape commonness and comparison-set size, both of which cancel in the ratio; and the model delivers the testable implications examined in the remainder of the paper—a placebo PDR of one with zero persistence, two-sided detection in which uniqueness is itself informative, persistence that strengthens as estimation noise falls, and a monotone link between recurrence strength and return predictability.

PDR Persistence

We test whether patterns with low PDR (strong recurrence) in one period continue to exhibit strong recurrence in future periods. For each segment $n \in \mathcal{D}_y$, we have both an in-sample PDR (PDR_n^{in}) and an out-of-sample PDR ($PDR_n^{out,h}$). We compute the Pearson and Spearman rank correlations between PDR_n^{in} and $PDR_n^{out,h}$ across all segments in year y. Under the null hypothesis that stock returns follow a random walk and segment recurrence is spurious, these correlations should be close to zero. A positive correlation indicates that the degree of pattern recurrence is a persistent characteristic that carries over to new data. We report both the pooled correlation (across all years) and year-by-year correlations to assess the consistency of the result.

PDR Quintile Transition Matrices

To further examine the stability of pattern recurrence, we construct transition matrices based on PDR quintiles. For each year y and horizon h, we sort all patterns in $\mathcal{D}_y$ into five quintiles based on their in-sample PDR_n^{in} (Q1 = lowest PDR / strongest recurrence, Q5 = highest PDR / weakest recurrence). We then sort the same patterns into quintiles based on their out-of-sample $PDR_n^{out,h}$. The transition matrix $\mathbf{T}^h$ has entries:

$$T^h_{q_1,q_2} = \Pr\left(\text{out-of-sample quintile} = q_2 \,\middle|\, \text{in-sample quintile} = q_1\right), \tag{7}$$

estimated as the fraction of patterns in in-sample quintile q_1 that fall into out-of-sample quintile q_2. Under the null hypothesis of no persistence, all entries of $\mathbf{T}^h$ should equal $1/5 = 0.20$. Diagonal entries exceeding 0.20 indicate that patterns tend to remain in the same recurrence quintile across time.

3 Pattern Recurrence in Stock Returns

This section presents the central empirical finding of the paper. Using the Matrix Profile-inspired exhaustive distance comparison described in Section 2.2, we show that stock price patterns exhibit genuine and persistent recurrence that is absent from random walk data. Section 3.1 documents that the degree of pattern recurrence, as measured by the PDR, is a stable characteristic that carries over from in-sample to out-of-sample periods. Section 3.2 reinforces this finding through transition matrices that track how segments move across PDR quintiles over time. Section 3.3 shows that the persistence is robust to alternative benchmark constructions, neighbor definitions, and segment lengths, and that it extends to international equity markets.

3.1 Pattern Distance Ratio (PDR) Persistence

We begin our empirical analysis by examining whether patterns exhibit persistent recurrence across time. Figure 2 and Table 1 present the persistence analysis of PDR across the one-, two-, three-, and five-year comparison-block specifications, compared to a random walk placebo test. In the placebo, we replace the real-to-real neighbor search with a real-to-random neighbor search using a fifth independent random walk series (Section 2.3), so that any apparent persistence can arise only from the methodology rather than from the data. Panel B of Table 1 shows the overall correlation between in-sample and out-of-sample PDR. The Pearson correlation increases monotonically across the block-length specifications, from 0.308 under one-year blocks to 0.486 under three-year and 0.603 under five-year blocks, with the Spearman rank correlation rising from 0.251 to 0.444. These positive and statistically significant correlations indicate that segments with low PDR (high recurrence) in one period tend to maintain high recurrence in subsequent periods. Because lengthening the block enlarges the comparison sample at the same time as it increases the temporal distance (Section 2.4), Panel D separates the two effects by holding the comparison blocks fixed at one year and varying only the horizon: the correlation is 0.308 one year apart, 0.275 two years apart, 0.270 three years apart, and 0.257 five years apart, with the year-by-year Spearman correlation positive in 53 of 54, 52 of 53, 49 of 52, and 50 of 50 block-pairs respectively.

In stark contrast, the random walk placebo test yields a correlation of 0.011 (Spearman 0.008)—roughly thirty times smaller than the real-data value and, on the decile plots of Figure 2, visually flat. The comparison that matters here is one of magnitude rather than of statistical significance, and deliberately so. The placebo is built from one realization of a random shuffle, so its persistence is itself a random variable; re-forming the placebo from the five independent shuffled series we generate yields twenty configurations whose mean correlations span only −0.003 to +0.017, but whose conventional cross-year t-statistics range from −0.5 to +3.5 purely with the draw. A significance test is the wrong instrument for this comparison in any case: with fifty-four year-pairs the standard error of the mean correlation is about 0.005, so a residual of no economic consequence is easily declared "significant," and failing to reject zero would not establish it. What the placebo can establish, and does under every configuration we examined, is that the methodology cannot manufacture persistence of the order the real data display: the largest placebo mean we obtain, 0.017, is one twentieth of the 0.317 measured on actual returns. Panel C examines year-by-year consistency and reveals that the mean Spearman correlation across individual years ranges from 0.233 (1-year) to 0.450 (5-year). The year-by-year Pearson correlation is positive in 100% of years at every real-data horizon. This universal positive persistence provides strong evidence against the pooled-i.i.d. shuffle null; Section 3.3 shows that the persistence also survives a far more demanding null that preserves stock-level distributional heterogeneity, volatility clustering, and factor comovement.

Figure 3 provides a time-series perspective on pattern persistence, with gray shaded areas indicating NBER recession periods. It plots the Pearson correlation between in-sample and out-of-sample PDR for each year from 1970 to 2023, revealing notable temporal variation with particularly strong persistence during the 1990s and relatively weaker persistence in the 2000s and 2010s. To examine whether pattern persistence varies with economic conditions, we compare recession periods (N=14 years) to expansion periods (N=40 years). The difference is small relative to year-to-year variation: the PDR correlation averages 0.272 during recessions versus 0.333 during expansions, a gap of 0.061 against a year-to-year standard deviation of 0.159. With fourteen recession years the comparison has little power, so we read it as showing no economically material regime dependence rather than as establishing its absence. This evidence suggests that pattern persistence is robust across economic cycles rather than driven by regime switching. The directional counterpart of this persistence—the rate at which a segment preserves the sign of its neighbor-outcome statistic from in-sample to out-of-sample—is reported by PDR quintile in Table 3 and analyzed in Section 4.

Several features of these results merit discussion. First, the fixed-block decomposition of Panel D shows that pattern recurrence is a slow-decaying characteristic: holding the comparison sample fixed, the correlation declines only from 0.308 at a one-year distance to 0.270 at three years and 0.257 at five years—retaining almost ninety percent of its one-year value at three years and about 83 percent at five—and remains positive in all but a handful of block-pairs. If recurrence were driven by transient factors such as temporary liquidity shocks or fleeting investor attention, the correlation would fall off sharply with distance; the observed decay is mild. The monotonic *rise* across the block-length columns of Panel B, by contrast, should not be read as persistence strengthening with distance: it reflects the growth of the comparison sample as the blocks lengthen—larger candidate sets reduce the estimation noise in each segment's nearest-neighbor distances, mechanically raising the correlation between the in-sample and out-of-sample estimates, exactly as Proposition 3(ii) of Appendix D predicts. Indeed, the two panels together bracket the phenomenon: temporal distance alone erodes the correlation slowly (Panel D), while enlarging the information set raises it (Panels B–C), and both movements are what a stable underlying recurrence characteristic measured with noise would produce.

Second, the contrast with the random walk placebo is sharp and economically informative. Persistence one to two orders of magnitude smaller under the placebo confirms that the methodology itself does not mechanically generate persistence of the magnitude we document: only when applied to real market data does persistence of order 0.3 emerge. This rules out the possibility that our results are artifacts of the distance metric, the z-normalization procedure, or the year-block comparison structure. The finding that 100% of individual years exhibit a positive Pearson correlation in real data—with a mean Spearman correlation of 0.233 and a year-to-year standard deviation of 0.124, so that the typical year sits nearly two standard deviations above the placebo's entire range—further underscores that pattern recurrence is a pervasive feature of stock markets, not confined to particular subperiods or market regimes.

Third, the absence of significant differences between recession and expansion periods suggests that the mechanisms generating recurrent price patterns operate continuously rather than being activated by specific economic conditions. This is consistent with the view that recurrence arises from persistent structural features of markets—such as the interaction of heterogeneous trading strategies, institutional rebalancing, or the gradual incorporation of information into prices—rather than from regime-specific phenomena like flight-to-quality or crisis-driven herding.

3.2 Transition Matrices and PDR Stability

Table 2 presents transition matrices showing the probability that a segment in PDR quintile q_1 (in-sample) moves to quintile q_2 (out-of-sample); the diagonal dominance and its strengthening with the horizon are immediately apparent. The diagonal elements measure the probability that segments remain in the same quintile across periods. For the 1-year horizon (Panel A), all five diagonal probabilities exceed the random walk expectation of 0.20, with the extreme quintiles standing well above the middle three: 0.269 (Q1) and 0.408 (Q5) against 0.229–0.248 for Q2–Q4. The diagonal dominance strengthens considerably as the comparison blocks lengthen: under the 3-year specification (Panel C), Q1 segments show 0.310 probability of remaining in Q1, while Q5 segments exhibit a remarkable 0.485 probability of staying in Q5.

The asymmetry between Q1 and Q5 is particularly noteworthy. Segments with the lowest PDR (Q1, highest recurrence) and highest PDR (Q5, lowest recurrence) both demonstrate stronger persistence than middle quintiles, suggesting that both ends of the PDR distribution represent stable pattern characteristics. In contrast, Panel D shows that the random walk benchmark produces nearly uniform transition probabilities around 0.20 across all cells, with no meaningful diagonal dominance. This stark difference confirms that PDR captures genuine and persistent pattern characteristics rather than transient noise.

The transition matrices offer several additional insights into the nature of pattern recurrence. First, the stronger diagonal persistence at Q5 (0.408–0.485) compared to Q1 (0.269–0.310) reveals an informative asymmetry: segments that are unique—in the sense of lacking close matches in the data—tend to remain unique over time. This is consistent with the interpretation that Q5 segments reflect idiosyncratic, firm-specific price dynamics that are inherently difficult for other stocks to replicate, whereas Q1 segments capture common price dynamics shared across many stocks. The fact that both extremes persist more strongly than the middle quintiles suggests that the PDR distribution has stable tails, with segments tending to sort into "highly recurrent" or "highly unique" categories that persist across years.

Second, the off-diagonal elements exhibit a clear gradient: the probability of transitioning to a distant quintile declines monotonically with the distance from the diagonal. For instance, in Panel A, a Q1 segment has a 23.4% chance of moving to Q2 but only a 13.3% chance of jumping to Q5. This smooth gradient indicates that PDR changes are typically incremental rather than abrupt, reinforcing the view that the degree of pattern recurrence is a stable, slowly evolving characteristic.

Third, the strengthening of diagonal dominance from the 1-year to the 3-year specification parallels the correlation results in Section 3.1—and, like them, partly reflects the growth of the comparison sample as the blocks lengthen (Section 2.4)—while providing a complementary, nonparametric confirmation: pattern recurrence is not merely correlated across time in a linear sense, but exhibits genuine rank-order stability. Together, the correlation analysis and transition matrices establish that PDR is a meaningful and durable property of stock price patterns—one that distinguishes real market dynamics from temporally unstructured data across all time horizons examined.

3.3 Robustness

The persistence documented in this section is robust to two natural concerns about how the PDR is constructed, summarized in Table A2. The first concern targets the benchmark. Because our random walk null shuffles returns across the cross-section within each year, it destroys each stock's own time-series structure; real 60-day segments could therefore appear systematically closer to one another than to the benchmark simply because of trivial univariate autocorrelation—such as volatility clustering—rather than genuine shape recurrence. To impose a far more demanding null, we reconstruct the benchmark using a stationary block bootstrap (mean block length of 10 trading days) applied to each stock's six-factor (FF5+UMD) residuals, leaving the stock's real factor exposures—and hence its systematic comovement and the bulk of its own autocorrelation—intact while scrambling only the idiosyncratic component. The second concern targets the neighbor set. A segment's 100 nearest neighbors may be dominated by contemporaneous, same-industry, same-size, and same-value peers, so that measured recurrence reflects common factor comovement rather than distinct pattern repetition. To rule this out, we impose a stricter neighbor-selection rule that partitions stocks into size $\times$ book-to-market $\times$ 48-industry $\times$ month cells and retains at most one neighbor per cell before taking the 100 nearest, forcing the neighbor set to be diverse across firms, sectors, and time.

Both alternatives leave the persistence result intact. Under the much harder block-bootstrap surrogate, the mean year-by-year Pearson correlation between in-sample and out-of-sample PDR is 0.247 (Spearman 0.176), positive in 52 of 53 year-pairs, with a cross-year t-statistic of 11.42 (Newey–West t = 7.63; Newey and West 1987). This is roughly 22% below the 0.317 baseline—the portion attributable to the trivial autocorrelation channel that the surrogate removes—but it remains large and overwhelmingly significant, and it survives further controlling for each segment's factor structure. Doubling the mean block length to 20 trading days—so that a 60-day window is stitched from roughly three blocks and within-window dependence at up to monthly scales, including slow volatility dynamics, is preserved in the null—changes nothing: the mean correlation is 0.262 (Newey–West t = 7.5), positive in 52 of 53 year-pairs, and the sign-preservation gradient across PDR quintiles is intact—the four most recurrent quintiles exceed the adjusted null by 2.2 to 2.9 percentage points against 1.7 for the most novel quintile (untabulated). The persistence that survives the surrogate is therefore not attributable to slow volatility structure either. Under the stricter neighbor rule, persistence is, if anything, marginally stronger than the baseline (mean Pearson 0.350 versus 0.317, positive in all 54 year-pairs), and pair by pair the stricter rule yields the higher correlation in every single year-pair (mean difference +0.033, $t = 14.6$). This confirms that the result is not an artifact of clustered, comoving neighbors. Both alternatives stand in sharp contrast to the random walk placebo, whose mean correlation of 0.014 is more than an order of magnitude smaller (Section 3.1).

The harder null also carries through to the downstream analyses that use the PDR, not merely to its persistence (Table A2, Panel B). Re-sorting segments on the surrogate-based PDR, the recurrent tail is, if anything, more stable than under the shuffle benchmark—the one-year Q1 retention probability rises from 0.269 to 0.362—while the unique tail moderates from 0.408 to 0.296, indicating that part of the baseline persistence of *uniqueness* reflects persistent stock-level distribution types that the surrogate absorbs into the null. More importantly, the link between recurrence and predictability survives: among high-$|T|$ segments, sign preservation still exceeds the adjusted null by 2.4 to 2.9 percentage points in the four most recurrent surrogate-PDR quintiles against 1.8 percentage points in the most novel quintile—the same ordering as under the shuffle benchmark, on a narrower gap (untabulated). The recurrence–predictability gradient of Section 4 is therefore not an artifact of distributional heterogeneity, volatility clustering, or factor comovement.

A third concern is that PDR persistence might simply restate the well-known persistence of volatility: volatile stocks trace unusual-looking paths, and volatility is itself highly persistent. Table A3 shows that it does not. Adding the formation-window idiosyncratic volatility to the persistence regression leaves the coefficient on in-sample PDR essentially unchanged, at 0.304 versus 0.317, and adding size, reversal, and the path's trend changes nothing further; idiosyncratic volatility enters significantly on its own but absorbs almost none of the persistence. Persistence is also present within every volatility tercile—a mean correlation of 0.215 among the least volatile segments, positive in 49 of 54 year-pairs, rising to 0.389 among the most volatile—so it is a property of trajectory shape rather than of scale. We conclude that PDR persistence reflects genuine idiosyncratic pattern recurrence rather than mechanical autocorrelation, factor comovement, or volatility persistence.

The persistence is also robust to alternative segment lengths. Using 30-day and 90-day specifications in addition to the 60-day baseline, the Pearson correlation between in-sample and out-of-sample PDR ranges from 0.287 to 0.333 (Table A4), with correlations positive in 98–100% of individual years (untabulated). It is likewise insensitive to the neighbor count: replacing $K = 100$ with $K = 50$ or $K = 200$ yields one-year pooled correlations of 0.354 and 0.235, respectively, each positive in all 54 year-pairs (untabulated). The corresponding sign-preservation robustness is reported alongside the predictability results in Section 4.

Finally, the persistence is not specific to U.S. equities. Replicating the PDR construction in the twelve international equity markets that both clear the cross-sectional depth requirement and can also be evaluated at Stage II (thirteen clear the depth rule; the United Kingdom is dropped because its Stage II series is too intermittent, as Table A5 explains), the in-sample/out-of-sample correlation is positive in every market—and in at least three quarters of the market-years within each—averaging 0.29 across markets (the cross-market average correlation carries $t = 8.5$; the average of the twelve individual market t-statistics is 7.8), as reported in Table A5; in four of the twelve markets the correlation exceeds the U.S. baseline, and the cross-market spread is wide, from 0.47 in Hong Kong to 0.09 in China. Section 5.6 returns to these twelve markets and deploys the full prediction engine in them as a one-shot out-of-sample test.

The requirement of "adequate cross-sectional depth" deserves a word, because it is what determines which markets can be studied at all and it binds far more tightly here than in the international asset-pricing literature. Studies that estimate factor exposures stock by stock—Chaieb et al. (2021) cover 47 countries—need only a long enough time series for each individual stock, so a market with a few dozen listed firms still contributes usable observations. The PDR is a fundamentally different object: it is a *local density* statistic, the distance from a segment to its nearest neighbors *within the same market and year*, normalized by the same quantity computed against a benchmark drawn from that market and year. Its precision therefore depends on how densely the contemporaneous cross-section populates the neighborhood of each trajectory, not on the length of any stock's history. In a thin market the nearest neighbors are far away by construction, the numerator and the denominator are both estimated with large error, and—exactly as Proposition 3(ii) of Appendix D predicts—the measured in-sample/out-of-sample correlation is attenuated toward zero by estimation noise rather than by any absence of structure.

We therefore require each market-year to contribute at least 5,000 segments observed in both the in-sample and the out-of-sample block, a rule fixed before any correlation was computed and applied mechanically. The rule is not binding at the margin for anything we report: every retained market clears the threshold in at least six market-years, Germany and Sweden being the least deep of the twelve. Six markets—Brazil, Switzerland, Ireland, Israel, Italy, and Poland—fell short of it on the original data and were not carried into the cleaned sample, and Singapore, which had cleared it beforehand, no longer does once the cleaning is applied. Retaining a thin market would not bias the results toward our hypothesis—noise attenuates the correlation—but it would add uninformative rows measured with error, and we prefer to state the depth requirement explicitly rather than average over it.

4 Recurrence Strength and Return Predictability

Having established that price patterns exhibit persistent recurrence (Section 3), we now examine whether the degree of recurrence is linked to the persistence of return predictability. We measure predictability using the T-statistic of each segment's nearest neighbors' future Fama–French 25-adjusted returns (formally defined in Appendix B). For each segment $n \in \mathcal{D}_y$, we compute both in-sample statistics (PDR_n^{in}, T_n^{in}) from the within-year comparison $\mathcal{D}_y$ vs. $\mathcal{D}_y$, and out-of-sample statistics ($PDR_n^{out,h}, T_n^{out,h}$) from $\mathcal{D}_y$ vs. $\mathcal{D}_{y+h}$ at horizons $h = 1, 2, 3$ years.

Table 3 reports sign preservation rates—the proportion of segments that maintain the same directional prediction (sign of T_n) from in-sample to out-of-sample. Among high-$|T|$ segments the rates reach 51.4–54.1% in the four most recurrent quintiles, well above the independence-adjusted null of approximately 50.2–50.4%, and decline toward the null in the most novel quintile. Across all four block lengths, sign preservation for Q1 (strongest recurrence) exceeds that for Q5 (weakest recurrence) by 1.6–2.4 percentage points, reinforcing the link between recurrence strength and the reliability of predictive signals. These results are robust to alternative segment lengths and holding periods: across all six combinations of pattern window (30, 60, and 90 days) and holding period (20 and 30 days), high-$|T|$ sign preservation reaches 50.9–53.6% in the four most recurrent quintiles and exceeds the independence-adjusted null in every such cell, and is strongest for the shortest windows and horizons (Table A6).

Because the neighbors' evaluation windows overlap within calendar months, the score's nominal scale overstates precision (Appendix B), and it is worth verifying that these conditioning results are not an artifact of ranking on correlated noise. Two checks confirm they are not. First, under the random walk placebo the same selection machinery produces correlations near zero (untabulated): extreme values of correlated noise do not recur in the next year's independent block. Second, recomputing the score with month-clustered standard errors—averaging neighbor outcomes within each evaluation month before forming the t-ratio across the roughly ten months represented—tightens the calibration (the share of segments crossing 1.96 falls from 10.4% to 7.7%) but leaves the substance intact: sign-preservation excesses over the adjusted null remain positive in every quintile and the most novel quintile remains the weakest, although the clustered score's lower precision—roughly ten effective observations in place of one hundred—attenuates the magnitude of the Q1–Q5 gradient (untabulated).

Together with the persistence evidence of Section 3, these results establish that the geometry of trajectory space captures economically meaningful regularities: the more strongly a pattern recurs, the more reliably its historical outcomes forecast future returns. Section 5 translates this segment-level link into a tradable long–short strategy and evaluates its risk-adjusted performance.

5 A Trajectory-Space Portfolio Strategy

We translate the pattern-based predictability documented in Section 4 into a tradable long–short strategy. The estimator is a direct implementation of the "location-predicts-return" mapping in the trajectory geometry: a trajectory's expected future return is estimated from the realized outcomes of historically similar trajectories. Section 5.1 describes the construction of the strategy, Section 5.2 reports its risk-adjusted performance, Section 5.3 verifies that the shape-similarity structure is what drives the prediction, Section 5.4 asks whether the resulting alpha is better read as a risk premium or as mispricing, and Section 5.5 measures the strategy's trading costs and shows that the gross alpha lives largely inside the transaction-cost band—the quantified limit to arbitrage behind its persistence. Finally, Section 5.6 transplants the engine, with every design choice frozen, to twelve international equity markets as a fully out-of-sample test.

5.1 Methodology

The estimator operates entirely in the trajectory space of Stage I: downstream, the model never sees raw returns—only the Stage I distance metric—so it is best understood as an approximate kernel ridge regression in the trajectory geometry, that is, a distance-weighted average of similar past outcomes.

The construction proceeds in four steps. First, we fix a set of *landmark* trajectories drawn from the *recurrence grid* of Stage I: from the segments of the trailing three years, we keep those whose neighbor-outcome score exceeds the conventional cutoff ($|t| \geq 1.96$) and draw 410 of them from each decile of that pool's PDR distribution, so that the grid is anchored at the recurrence structures identified in Stage I and spans the recurrent and the novel regions alike. The draw gives up to 4,100 landmarks per test year—4,100 in forty-nine of the fifty-one test years, and 1,612 and 3,239 in the two earliest, where the pool is thin. Five independent draws are taken and their predictions averaged by within-month rank, so that no single draw determines the signal. Second, each 60-day standardized return trajectory is represented by the vector of its Gaussian-kernel similarities to the landmark trajectories, which serves as a Nyström / random-feature approximation of the full kernel (Williams and Seeger, 2001; Rahimi and Recht, 2008). Third, we predict the FF25-adjusted forward return (Appendix B), residualized each month on the segment's five- and 21-day reversal characteristics, by ridge regression on this similarity vector, estimated on an expanding, purged window with a one-month embargo between the training sample and the prediction date. Fourth, following Da et al. (2014), the fitted score is orthogonalized each month to the same two characteristics, and the residual is the trading signal used throughout. Retaining the full high-dimensional similarity vector and regularizing with ridge, rather than selecting a few features, follows the virtue-of-complexity logic for return prediction (Kelly et al., 2024).

One clarification about the role of the landmarks is essential for interpreting the results that follow. The landmarks are a computational basis, not the source of the forecast: the information the estimator exploits resides entirely in the training pairs of trajectories and realized outcomes, and the landmark set only determines the finite-dimensional basis in which the kernel regression is expressed. Under the framework of Appendix D, any landmark set whose coverage includes the dense regions of trajectory space spans nearly the same space of smooth functions of location (Proposition 4(iii) and Remark 2), so the estimator's performance should be—and, as the design perturbations reported in Section 5.2 confirm, is—nearly invariant to how the landmarks are selected. In particular, landmarks drawn at random from the pool of real trajectories are not a recurrence-free alternative: random draws from real data land in the dense regions of shape space in proportion to their density, and that density structure is precisely the recurrence documented in Section 3. We use the recurrence grid because it is a natural, pre-specified way to fix the basis, but the evidence connecting recurrence to the predictability does not run through the landmark choice; it rests on the landmark-independent results of Sections 3, 4, and 5.3.

Nagel (2025) cautions that this logic can be illusory: when a high-dimensional kernel regression is trained on a short rolling window, similarity in the predictor space mainly reflects temporal proximity, so the forecast reduces to a recency-weighted average of recent returns—an implicit momentum strategy dressed up as complexity. Three features of our design rule out this degeneracy. First, we forecast the cross-section rather than the aggregate market: stocks are ranked within each month, so any common recency component nets out of the long–short position. Second, the training window is expanding—spanning decades, with equal weight on all history—rather than short and rolling. The predictive map from trajectory shape to future return is long-memory stable, not a repackaging of recent returns. Third, the scrambled-target placebo of Section 5.3 breaks the correspondence between shapes and outcomes while leaving the entire kernel similarity structure intact, and predictability vanishes—so the forecast's content resides in the shape–outcome map rather than in mechanical properties of kernel weighting.

At the end of each month, we sort stocks into quintiles by the predicted return and trade a long–short portfolio that buys the top quintile (Q5) and sells the bottom quintile (Q1), under both equal and value weighting, over the out-of-sample period 1975–2025.

5.2 Performance

Table 4 reports the strategy's long–short alphas across factor models. The alpha is large and highly significant under every model, for both equal- and value-weighted portfolios (FF6 alphas of 8.39% and 5.92% per year, respectively). Because the engine reads a sixty-day price path whose right edge is the most recent week, the natural question is how much of this is short-horizon reversal (Jegadeesh, 1990; Lehmann, 1990). Following Da et al. (2014), the signal is orthogonalized each month to the segment's own five-day and 21-day reversal characteristics, so the answer is built in: the long–short portfolio's loadings on French's published *monthly* reversal factor and on a *weekly* reversal factor built inside our own universe under the strategy's own rule are small (0.03, $t = 1.03$, and -0.05, $t = -1.60$, value-weighted, over the full sample), and the alpha survives adding either factor (FF6 + ST-Rev 8.54% equal-weighted and 5.81% value-weighted; with WST-Rev as well, 7.30% and 6.33%).

Because a quintile sort discards the interior of the cross-section, the portfolio alpha is a low-power summary of the signal in a short window; the monthly rank information coefficient, which uses every stock and is the accuracy measure employed throughout Section 5.3, is 0.026 ($t = 5.5$) as an unweighted Spearman correlation and 0.040 ($t = 4.3$) when the rank correlation is weighted by market capitalization—the two versions we refer to below as equal- and value-weighted—over 2009–2025. The strongest recent-period evidence, however, is out-of-sample by construction: transplanted with every design choice frozen, the engine's global country-average portfolio earns a significant alpha across twelve international markets over 2004–2025 (Section 5.6). Cumulative growth is steady: \$1 invested in the long–short portfolio grows to roughly \$94 equal-weighted and \$22 value-weighted over 1975–2025.

The alpha is also robust to the estimator's design choices (untabulated): against the main specification's 8.39% per year equal-weighted and 5.92% value-weighted, a two-month embargo well beyond the 60-day formation overlap leaves 8.32% ($t = 9.3$) and 6.11% ($t = 5.0$), and replacing the recurrence grid with random landmarks leaves 8.27% ($t = 9.4$) and 6.24% ($t = 5.0$); both remain significant after 2009 as well—indicating that the result is genuinely out-of-sample rather than an artifact of any single design choice. The last of these perturbations deserves emphasis, because its interpretation is easily inverted. The near-invariance of the alpha to the landmark selection does not say that recurrence is irrelevant to the predictability; it says that the landmark set is what the framework says it is—an interchangeable basis (Section 5.1; Appendix D, Remark 2). Landmarks drawn at random from real trajectories inherit the density structure of the real data, so both bases cover the same recurrent regions of shape space, and the alpha is a property of the trajectory–outcome mapping itself rather than of any particular landmark rule. Whether recurrence underlies that mapping is a question about the data, not about the basis, and it is taken up on its own terms in Section 5.3. The simpler specification that was fixed before any of these refinements were examined earns 7.9% per year equal-weighted ($t = 9.2$) and 5.8% value-weighted ($t = 4.7$) over the full sample and remains significant after 2009 (8.3%, $t = 5.1$, and 8.5%, $t = 4.1$), so the finding does not rest on the refinements.

A remaining concern is that the estimator was chosen after examining the data, so that the reported t-statistics overstate the evidence. The out-of-sample label deserves precision here, because it can mean two different things. In the statistical sense—no forecast uses information realized after its formation date—the design is strictly recursive: expanding training windows, purged and embargoed, with landmarks drawn point-in-time. In the stronger, design-level sense—that no specification choice was made with knowledge of test-period performance—the relevant question is how much room such choices leave, and it can be bounded. First, several apparent degrees of freedom are not free: the kernel bandwidth is set by the median heuristic and the ridge penalty by generalized cross-validation—data-driven rules fixed ex ante, not tuned to performance. Second, the choices that are free turn out not to matter: across the design perturbations just described, the full-sample equal-weighted alpha varies only between roughly 7.9% and 8.4% per year. The design space is a plateau, not a spike—and a snooped result lives on a spike. Third, the specification fixed before any of the refinements were examined (the pre-committed ridge-on-4096 above) already earns 7.9% per year ($t = 9.2$), so the headline does not rest on any searched choice. Finally, we ask directly how large a t-statistic this pipeline produces when there is nothing to find. Holding fixed everything except the object of interest—the trajectories, the recurrence-grid landmark selection, the kernel, the five landmark draws, the ridge/GCV readout, the reversal orthogonalization, the expanding-window purge, and the portfolio rule—we permute the training target within each calendar month, which destroys the correspondence between a segment's shape and its realized outcome while preserving the monthly cross-sectional distribution of the target, and re-estimate the entire pipeline. Repeating this $K = 100$ times yields the empirical null distribution of the long–short FF6 alpha t-statistic that the pipeline generates by construction; the exercise is deliberately conservative, in that landmark selection is left at its real values, so the null retains a basis built from genuine recurrence information. The null is centered on zero but wide—its 99th percentile reaches 4.7 equal-weighted and 3.3 value-weighted over the full sample—confirming that a search of this kind can manufacture conventionally significant t-statistics. The actual estimates nonetheless lie outside the entire null distribution in every cell: the FF6 t-statistics of Table 4 are 9.38 equal-weighted against a null maximum of 5.31 and 4.82 value-weighted against 3.59 over the full sample, and, in the post-2009 subsample where the evidence is weakest, 4.98 against 2.81 and 4.03 against 2.82. The randomization p-value is 0.01 in every cell.

Two further questions concern what the signal is rather than whether it is real. First, since the engine reads price paths, it may be a repackaging of the technical rules and characteristics already known to forecast returns. Table 6 runs the horse race directly, in the format of Han et al. (2016): every regressor is winsorized and standardized within the month, so slopes are comparable across variables, and the regressions are run both equal- and value-weighted. Moving-average rules built from the same 60-day window do predict returns in our sample, and so do reversal, momentum, idiosyncratic volatility, size, and liquidity; but the signal's Fama–MacBeth coefficient falls only from 0.238 to 0.152 equal-weighted when all of them are included simultaneously, and its t-statistic remains 8.2. Adding the weekly reversal characteristic—to which the signal is already orthogonalized within each month—leaves the coefficient undiminished (0.182, $t = 8.3$), while that column absorbs much of the moving-average rules' explanatory power (the price-versus-average coefficient falls from −1.011 to −0.408), which suggests that what such rules read out of a 60-day window is in part short-horizon reversal. Value-weighted (Table A7), the coefficient falls from 0.175 to 0.076 ($t = 3.1$) with every control included, so the signal's incremental content is concentrated in smaller stocks. Roughly two-thirds of the signal's equal-weighted predictive content is orthogonal to the technical-rule and characteristic space. Second, because the alpha is larger in small, high-spread stocks, it could reflect a microcap premium. It does not (untabulated): excluding stocks below the NYSE 20th size percentile, or restricting the universe to NYSE listings, leaves alphas of 6.3–6.9% equal-weighted and 4.7–5.5% value-weighted, significant in every window, 2009 onward included.

5.3 Recurrence and Signal Accuracy

The strategy's premise is that a pattern's future return is predicted by the realized outcomes of historically similar patterns. Because the landmark basis is interchangeable (Section 5.2), the link between the recurrence structure of Stage I and the strategy's performance must be established on evidence that does not run through the landmark choice. Two such pieces of evidence appear in this paper: the Stage I gradient linking recurrence strength to the persistence of neighbor predictability (Table 3, and Section 3.3 under the block-bootstrap surrogate); and the accuracy gradient and placebo documented in this subsection. Here, two tests confirm that the similarity structure is what drives the prediction. First, the signal is least accurate for novel patterns. Sorting each month on PDR—a low PDR meaning the shape has closer historical analogs relative to a random walk—the cross-sectional rank information coefficient is lowest for novel patterns and higher in the two more-recurrent terciles: the novel tercile has the lowest rank-IC in the full sample under both equal- and value-weighting, and the recurrent−novel gap is positive in the full sample (+0.58 percentage points equal-weighted, +0.49 value-weighted) and after 2000 (+0.63

equal-weighted, +1.33 value-weighted). The two weightings part company in the last window: after 2009 the equal-weighted gap closes to −0.02 percentage points while the value-weighted gap widens to +2.39 (untabulated). We report a rank-IC rather than a long–short alpha here because the alpha conflates signal accuracy with return dispersion—which is itself larger among novel patterns—whereas the scale-free rank-IC isolates accuracy; a Fama–MacBeth (Fama and MacBeth, 1973) interaction of the signal with recurrence is positive but no longer significant at conventional levels ($t = 1.89$ in the full sample). If anything, this sort understates the gradient: the PDR used here is estimated against a trailing three-year comparison window, whereas the engine's training set spans decades, so a shape whose recent recurrence is modest may still be well represented in the full training history—the monthly recurrence sort is a noisy proxy for the recurrence the estimator actually exploits. This mirrors, in the cross-section of individual stocks, the aggregate "familiarity" result of Chen et al. (2026), who find that news-based historical-analog predictions of the market are stronger when the current economic state more closely rhymes with history. Second, a placebo that scrambles the correspondence between a pattern's shape and its realized outcome—permuting the training targets within each month, so the model is fit on mismatched shapes—destroys predictability entirely: the cross-sectional rank-IC collapses from 0.027 ($t = 11.4$) to −0.001 ($t = -0.5$) equal-weighted and from 0.023 ($t = 4.6$) to 0.002 ($t = 0.5$) value-weighted in the full sample, and the scrambled signal is insignificant in every window after 2000 and after 2009 as well (untabulated). The predictability is thus a genuine property of the shape-similarity mapping rather than a mechanical artifact.

Two further tests sharpen the link by targeting, respectively, the conditioning variable and the training information itself (both untabulated). First, we re-measure recurrence against the estimator's *own* training history. For every segment in test year Y we compute the mean z-normalized distance to its $K = 100$ nearest neighbors in the engine's full training pool (all segments with year $< Y$, under the same purge), and divide it by the same quantity computed against a random-walk counterpart pool of the same historical years—a "historical PDR" that asks how much more closely the shape is matched in the actual training data than chance would imply. Sorting each month on this measure, the signal's accuracy falls monotonically as a shape's historical recurrence weakens: the equal-weighted rank-IC is 0.030 for the densest tercile, 0.028 for the middle, and 0.022 for the sparsest tercile in the full sample, and a Fama–MacBeth interaction of the signal with historical recurrence carries a t-statistic of 2.94 (3.18

after 2000). The normalization is doing exactly the work the framework assigns it: sorting on the *raw* distance to the training support produces no such gradient (if anything the ordering reverses)—raw proximity conflates recurrence with generic shape commonness, as in Remark 1 of Appendix D—whereas the chance-normalized measure isolates the recurrence component that conditions accuracy. One further decomposition is informative. When the denominator is instead the block-bootstrap surrogate of Section 3.3—so that own-stock dependence, distribution type, and factor comovement are absorbed into the null—the residual cross-sectional variation in historical recurrence shrinks by half and the engine's accuracy gradient flattens (untabulated). We read the two normalizations together descriptively: the engine's accuracy differences line up with recurrence measured against temporally unstructured data, part of which reflects the own-stock and common structure that the surrogate reclassifies as null, whereas the segment-level predictability gradient of Section 4 survives even the surrogate normalization (Section 3.3).

Second, we move the information rather than the basis. Holding the landmark grid, the target construction, and every other design choice fixed, we refit the ridge on restricted training sets: only the densest 30% of training segments (lowest PDR within their pattern year), only the sparsest 30% (highest PDR), and—as a size-matched control—a uniform random 30%. The outcome is symmetric with the landmark result, and we report it in the same spirit of transparency: the map is recoverable from any slice. Training on the densest 30% performs no better than training on a random 30% (the paired monthly difference in rank-IC is -0.001, $t = -0.7$), and all four training sets—including one restricted to the sparsest segments—produce highly correlated signals with statistically indistinguishable portfolio alphas; the sparse-trained variant is, if anything, marginally more accurate (+0.002 rank-IC over the random control, $t = 1.0$), consistent with recurrent segments being informationally redundant—near-duplicates of one another—so that novel segments contribute more independent coverage per observation. Under the framework this is the expected behavior of a rank-based estimator: restricting the training set changes the share of template draws and hence attenuates the *scale* of the fitted map, but a uniformly attenuated map induces the same cross-sectional ranking (Appendix D, Proposition 4(iii)). The two experiments together sharpen what the recurrence link is and is not: the shape–outcome map is a *global* property of the trajectory space, learnable from any sufficiently large sample of it—which is why neither the landmark basis nor the training subset matters—while *where the map is reliable* is governed by recurrence at the point of prediction, as the historical-PDR gradient above and the Stage I gradient of Table 3 both show. The map is global; its reliability is local, and recurrence is what indexes it.

5.4 Interpretation: Risk Premium or Mispricing?

The predictability could in principle reflect compensation for a priced risk or a mispricing that arbitrage competes away. The two readings make different empirical predictions: a risk premium should be roughly invariant to the cost of arbitrage and need not weaken as arbitrage capital enters, whereas a mispricing should concentrate in hard-to-arbitrage securities and should weaken as that capital competes it away (Shleifer and Vishny, 1997; McLean and Pontiff, 2016; Stambaugh et al., 2015). Because a raw long–short alpha is mechanically larger where returns are more dispersed, we read each conditional alpha alongside two scale-free measures—the rank information coefficient and the Sharpe ratio—that isolate signal quality from dispersion (Table 5). Two pieces of evidence favor the mispricing reading. First, the predictability is somewhat stronger in small stocks: the small-cap tercile has a larger equal-weighted alpha and a higher equal-weighted Sharpe ratio than the large-cap tercile in every window, and a modestly higher rank-IC in the full sample and after 2000 (Table 5: rank-IC 0.028 versus 0.021 over the full sample and 0.025 versus 0.023 after 2000; equal-weighted Sharpe ratios 1.51 versus 0.67 and 1.25 versus 0.65). The rank-IC gradient is modest even though the Sharpe gradient is not, so we read it as consistent with, rather than decisive evidence for, limits to arbitrage binding most among small firms (Shleifer and Vishny, 1997). The same conclusion emerges when we sort directly on the price of arbitrage itself: forming portfolios within terciles of the estimated effective spread (the stock-level trading-cost measure constructed in Section 5.5), the rank-IC rises monotonically with the spread, in the full sample and after 2000 alike (0.015, 0.022, and 0.035 from the low- to the high-spread tercile over the full sample; 0.015, 0.018, and 0.032 after 2000), and so do the equal-weighted alpha, the value-weighted alpha and the Sharpe ratios; the signal is thus most accurate in exactly the stocks where trading it is most expensive (Novy-Marx and Velikov, 2016) (untabulated). Second, the short leg—overpriced names that are costly to short—carries most of the full-sample spread (−6.0% of the 8.4% equal-weighted alpha, −5.4% of 5.9% value-weighted), and it is the part of the position that fades. Its magnitude falls by roughly a quarter equal-weighted after 2000 and keeps falling thereafter: by 2009–2025 the short leg is −3.3% equal-weighted ($t = -1.9$), no longer distinguishable from zero at the 5% level, and −3.7% value-weighted ($t = -2.2$), while the long leg strengthens from 2.4% ($t = 2.6$) over the full sample to 5.1% ($t = 3.1$) and 4.4% ($t = 2.9$) (untabulated). This is what one expects if the overpricing was competed away as arbitrage capital expanded. One commonly used cut, by contrast, proves uninformative here: sorting on idiosyncratic volatility produces no gradient in signal quality—the equal-weighted alpha rises monotonically across volatility terciles in the full sample (4.3%, 7.5% and 12.2% from low to high) while the rank-IC is essentially flat (0.024, 0.027, and 0.027), which is what one expects when an alpha gradient reflects return dispersion rather than accuracy—so a volatility sort cannot separate dispersion from predictability here (Ang et al., 2006; Stambaugh et al., 2015), and we do not rely on it (untabulated). Taken together, the trading-cost gradient, the size gradient, and the post-2000 attenuation of the short-leg correction are more naturally read as a mispricing competed away by arbitrage than as a stable risk premium. We emphasize, however, that this evidence is suggestive rather than decisive. By the joint-hypothesis problem (Fama, 1970), a risk-adjusted alpha is always a joint test of market efficiency and of the assumed factor model, so we cannot rule out that the predictability compensates for exposure to a priced risk that our factor models do not capture—each piece of our evidence is consistent with limits to arbitrage but does not, on its own, exclude a risk-based account. We therefore read the results as favoring a mispricing interpretation without claiming to settle the question.

Under this reading, the residual that survives—significant alpha in large, liquid names after 2000 (after 2009, at the 5% level only value-weighted)—is consistent with the binding limit to arbitrage here being *discovery cost*: the signal is recoverable only through the full geometric engine (simple neighbor or cluster expressions die out of sample), so the computational complexity of extraction is itself what preserves it. This is consistent with the virtue-of-complexity logic (Kelly et al., 2024) and with the view that the robustness of market efficiency partly reflects the difficulty of distilling tradable signals from low signal-to-noise data.

5.5 Implementability and the Cost of Arbitrage

The alphas reported above are gross of trading costs, and the signal turns over quickly: the strategy replaces roughly 80% of each leg every month under monthly rebalancing. This subsection asks what survives realistic transaction costs, using the accounting standard of the anomaly trading-cost literature (Novy-Marx and Velikov, 2016; Chen and Velikov, 2023). For every stock and month we estimate the effective bid–ask spread from daily high, low, and closing prices as the average of the Corwin and Schultz (2012) and Abdi and Ranaldo (2017) estimators (the two are highly correlated in our data, with cross-sectional correlation 0.95), and we charge each trade at the stock's estimated half-spread in the month it is executed. Turnover is computed after drifting the previous month's weights by their realized holding-period returns, so passive appreciation is not counted as trading. Alongside the baseline monthly-rebalanced portfolio we report a cost-mitigated version in the spirit of Novy-Marx and Velikov (2016): a buy/hold band that enters a position only when a stock reaches the extreme quintile of the predicted-return ranking but holds it until it leaves the extreme two quintiles, which cuts turnover by roughly one fifth to one quarter while lowering the gross alpha only modestly.

Table 7 reports the result in the form Grundy and Martin (2001) give it: what the strategy earns, how much it trades, and the trading cost that would exhaust the earnings. We report the break-even cost rather than a net-of-cost alpha because it is the statistic that does not depend on how well spreads can be measured—it uses only the alpha and the turnover—and because the reader's own execution cost, not ours, is what decides implementability (Bajgrowicz and Scaillet, 2012). At a turnover of roughly 82% of each leg per month the strategy trades about 3.3 dollars a month for every dollar held on a side, so an alpha of 8% per year is exhausted by a one-way cost of 21.3 basis points equal-weighted and 14.6 value-weighted. The buy/hold band cuts turnover by a quarter and raises the hurdle to 17–24 basis points; the band's figures, here and below, are untabulated. Restricting both legs to the largest NYSE size tercile lowers the hurdle rather than raising it—the gross alpha falls while the turnover does not—to 11–18 basis points under monthly rebalancing and 14–23 basis points with the band. These hurdles remain demanding: they are of the order of the effective half-spreads that large, liquid stocks have carried since decimalization, and far below those of small stocks in any period. The second column applies the stricter of Grundy and Martin's two conventions—the cost at which the profit can no longer be distinguished from zero rather than the cost at which it vanishes—and is naturally lower, 9–17 basis points for the monthly-rebalanced portfolio. It falls to 7.3 basis points among the largest firms after 2009 value-weighted, and to 2.3 equal-weighted: in those cells the evidence rather than the profit is what costs would erase first.

Our own estimates of what the strategy paid put it on the wrong side of that line, though we do not rest the conclusion on them. Charging every trade at the stock's estimated effective half-spread—the average of the Corwin and Schultz (2012) and Abdi and Ranaldo (2017) high–low estimators, which average 32–48 basis points across the windows—the cost drain is 13% to 18% per year, about twice the gross alpha, and the net alpha is negative in every window, weighting and panel; even the cost-mitigated strategy loses −6.9% per year equal-weighted ($t = -8.5$) and −5.1% value-weighted ($t = -4.7$) over the full sample (untabulated). Two considerations qualify those figures, both in the same direction. Daily high–low estimators overstate effective spreads for liquid stocks after decimalization (Chen and Velikov, 2023, Figure 2)—our own estimates for large firms after 2009, 30–33 basis points, are an order of magnitude above what intraday data imply for such stocks—and realized institutional costs are lower than academic spread estimates (Frazzini et al., 2018). A trader whose costs are in the single digits of basis points would clear the 13–17 basis-point hurdle of the post-2009 large-cap portfolio, or the 16–20 basis points it requires with the band. The reading is therefore two-sided, and deliberately so. Before decimalization, when even large stocks carried half-spreads well above the strategy's hurdle, these profits were not capturable, and the strategy's gross return tracks the estimated cost of trading it over five decades (below); in the modern large-cap universe the answer turns on execution quality. At no point in five decades did the strategy offer a large free lunch to the marginal arbitrageur.

Effective spreads price the marginal trade; price impact asks how much capital the opportunity could absorb. We model impact with the square-root law implied by market microstructure invariance (Kyle and Obizhaeva, 2016) and consistent with realized institutional costs (Frazzini et al., 2018): a trade in stock i costs $\kappa\, \sigma_i \sqrt{q_i}$ per dollar traded, where σ_i is the stock's daily return volatility, q_i is the trade's share of daily dollar volume (Nasdaq volume adjusted for inter-dealer double counting following Gao and Ritter (2010)), $\kappa = 0.5$, and orders are worked over ten trading days. Because the impact drag scales with the square root of assets under management, the fund size at which impact alone consumes the gross alpha—an upper bound on capacity, since it ignores spread costs entirely—is well defined (untabulated). The bound is strikingly small exactly where the alpha is large. The equal-weighted strategy, whose full-sample gross alpha is 8.39% per year, could absorb only about \$84 million before impact alone erased it, and at \$100 million more than a quarter of its trade dollars would already exceed a 5% daily participation rate. The value-weighted strategy in the modern sample could in principle absorb several billion dollars (roughly \$14.6 billion over 2009–2025)—but there the marginal economics have already failed, since estimated spread costs alone exceed the gross alpha. The two constraints are complementary and never simultaneously loose: when the margin was wide—in small stocks and early decades—the scale was negligible; by the time scale became available in large liquid names, the margin was gone. Halving κ quadruples the capacity bounds without changing this conclusion.

Far from undermining the paper's findings, this is the piece that completes them. The five-year rolling gross long–short return tracks the five-year rolling cost of trading it closely (correlation 0.67–0.68, untabulated), with the gross return lying below the cost bound for almost the whole sample, reaching it in only three months equal-weighted, all in 2001, and in twenty-two value-weighted, all but one of them between late 2000 and 2003. This is the Grossman and Stiglitz (1980) equilibrium in a single comparison: the mispricing documented in Section 5.4 persists *inside* the no-arbitrage band that transaction costs create. The market is efficient in the net-of-costs sense of Fama (1991); what our geometric representation recovers is the structure of prices *within* the band, which is real, persistent, and predictive, but bounded by the very frictions that allow it to survive. The predictability documented in this paper should therefore be read as evidence about how prices form, not as a claim of implementable arbitrage profits.

5.6 Out-of-Sample Evidence from International Markets

The most demanding test of a return predictor estimated on fifty years of U.S. data is to deploy it, unchanged, somewhere it has never been. We transplant the engine to the twelve international equity markets of Table A5—the markets in which Section 3.3 documented that the recurrence structure replicates—using daily Compustat Global data cleaned as described there (that table also reports the sample construction: segments, stocks per month, and each market's out-of-sample window). These screens play the role that the Ince and Porter (2006) and Hou et al. (2011) filters play in Datastream-based international studies: most of the pathologies those filters target—non-common securities, padded post-delisting returns—do not arise in Compustat Global once the issue-type screen is applied, and the file retains inactive securities, so the sample is free of survivorship bias. Returns are in local currency; because both legs of each long–short portfolio are denominated in the same currency, exchange-rate movements affect the spread only at second order, and converting every country's spread to U.S. dollars changes the global composite alpha by less than one basis point per month. The test design follows the conventions of the international anomaly-transplant literature (McLean et al., 2009; Watanabe et al., 2013; Jacobs and Müller, 2020): within-market portfolio construction, regional factor adjustment (Fama and

French, 2012), and pooled cross-country regressions on within-country ranks. Every design choice is frozen at its U.S. value: the recurrence-grid landmark rule, the median-bandwidth kernel, the ridge-penalty selection rule, the purged expanding window, the rank-based target, and the portfolio rules. To be precise about what "frozen" means: the rules and hyperparameters are fixed, while the objects they produce—each market's landmark grid, bandwidth, penalty, and regression coefficients—are re-estimated from that market's own data, exactly as a researcher deploying the U.S. specification abroad would do. Each market is evaluated exactly once. The claim that the specification predates the international data construction is verifiable rather than merely asserted: the engine's specification was finalized before the first Compustat Global download, and timestamped code, together with a replication package, is available from the authors. This is therefore a genuine one-shot out-of-sample test: no design choice was made with knowledge of any result reported below. Out-of-sample predictions begin between 1995 and 2013, depending on when a market's cross-section supports the minimum training sample, and run through December 2025. Within each market we evaluate the monthly quintile long–short portfolio against the corresponding regional five-factor-plus-momentum model (Fama and French, 2017), taking the factor series from Kenneth French's data library: the Europe, Japan, and Asia Pacific ex Japan series for the developed markets and the library's Emerging Markets series for China, India, South Korea, Malaysia, Taiwan, and Thailand. Benchmarking regional portfolios against local rather than global factors follows Fama and French (2012), who show that global models fail to price regional portfolios; the emerging-market series is distributed by the same library and constructed by the same procedure, but lies outside the developed-market samples of both papers. No international short-term reversal factor is available.

Table 8 reports the results. Three layers of evidence emerge. First, the sign pattern: the raw equal-weighted long–short return is positive in all twelve markets (the per-market raw returns are untabulated; Table 8 reports the factor-adjusted values) and remains positive in all twelve under regional factor adjustment, while the raw value-weighted return is positive in eleven of twelve; under the null of no effect, twelve positive signs out of twelve occur with probability $2^{-12} \approx 0.02\%$. The equal-weighted alpha is individually significant in ten of the twelve markets; the two exceptions, China and Germany, are positive but not distinguishable from zero (3.24% and 2.98% per year, $t = 1.63$ and $t = 1.23$). Second, the global series, constructed as the *country-average portfolio* of Chui et al. (2010), which places equal weight on each country's own long–short portfolio (Asness et al. 2013 aggregate the same way but weight each market by the inverse of the return volatility of that market's passive benchmark index). We follow their terminology deliberately: in Chui et al. (2010) the *composite* portfolio is the contrasting object that pools stocks across markets, which we report below as a robustness check. One element of our implementation is ours rather than theirs, and we flag it as such: we require at least five active markets in a month, where they require two. An equal-weighted average of the country long–short returns earns 6.34% per year equal-weighted ($t = 10.8$) and 4.97% value-weighted ($t = 5.7$) over 2004–2025. Following Chui et al. (2010) and Jacobs and Müller (2020), we treat these raw figures as the primary statistic for the aggregate: French's Developed factors do not span the six markets in our set that belong to his Emerging universe, so a regional-factor adjustment of the aggregate is less well posed than it is market by market. It changes little in any case—the Developed-factor alpha is 0.52% per month ($t = 10.3$) equal-weighted and 0.44% ($t = 5.9$) value-weighted (untabulated). Splitting the portfolio so that each half faces a model that does span it settles the question directly. The six markets in French's Developed universe earn 6.82% per year against the Developed factors ($t = 7.7$) equal-weighted and 4.78% ($t = 3.5$) value-weighted; the six in his Emerging universe earn 5.96% against the Emerging factors ($t = 7.1$) equal-weighted and 5.13% ($t = 4.0$) value-weighted. The result therefore holds on both sides of the split under both weightings.

As in the United States, the signal is orthogonalized within each market-month to the five- and 21-day reversal characteristics before the sort, and every figure above is for the orthogonalized signal. The series draws its power from breadth rather than from any single market—the average pairwise correlation between country long–short series is only 0.02 equal-weighted and 0.01 value-weighted, so the country-average series has a monthly volatility of 0.79% against a median of 2.10% for the individual markets, and the leave-one-out t-statistics range from 5.1 to 6.5 (value-weighted). The result is robust to how the markets are combined: converting each country's spread to U.S. dollars leaves the alpha essentially unchanged (6.2% per year, $t = 10.2$). Jacobs and Müller (2020) argue that no single aggregation of anomaly returns across countries is optimal—the choice is pooled versus composite across markets and equal- versus value-weighted within them—and compute all four combinations; we do the same. The *composite* portfolio of Chui et al. (2010)—which instead merges every market's extreme-quintile stocks into single global legs, so that larger markets carry more weight—earns 4.8% per year equal-weighted

(t = 7.9); its value-weighted counterpart, dominated by the largest names of the largest markets, is positive but insignificant (1.6% per year, t = 0.87), consistent with the signal being weakest among mega-caps. Third, a pooled cross-country Fama–MacBeth regression (Panel B): with all variables expressed as within-country-month percentile ranks, the signal carries a slope of 0.53% per month (t = 7.4) univariately and 0.53% (t = 7.8) after controlling for size, book-to-market, prior-month reversal, and 12–2 momentum—the control set of McLean et al. (2009) and Watanabe et al. (2013), the two studies that run the same kind of pooled stock-level regression across markets. The wrong-horizon problem documented in the U.S. repeats abroad, and the international data make it unusually clear. Adding the last-week return drives the prior-month slope from −0.51 (t = −2.43) to −0.08 (t = −0.42) while the weekly term itself carries −1.07 (t = −6.25): the monthly control that this literature uses was standing in for a weekly effect. The signal survives the correction in every specification: 0.55% (t = 8.0) with the last week controlled. Book-to-market is itself priced in this sample (0.68, t = 3.1), so the value effect is present and the signal simply does not overlap it. The international predictability is therefore not a repackaging of known characteristics.

Two features of the international sample bear noting. Beyond the Chaieb et al. (2021) screens it applies a $10 million market-capitalization floor. This floor is our own addition rather than part of their protocol; it plays the role that the NYSE size percentile and the $5 price screen play in the U.S. sample, but it is an absolute threshold and a far weaker one, so the large equal-weighted magnitudes in some markets (Australia, Sweden) are concentrated in small firms. Restricting each market to the largest firms covering 90% of aggregate capitalization—the big-stock breakpoint of Fama and French (2012), applied here within each country-month rather than within a region at each June—deflates exactly those cells—the Swedish equal-weighted alpha falls from 9.49% to 2.39% and the Australian from 10.02% to 4.54% per year—but the global country-average series survives in the large-cap universe: its Developed-factor alpha is 4.76% per year equal-weighted (t = 6.1) and 4.76% value-weighted (t = 4.8; untabulated). And the out-of-sample windows vary widely in length (from Thailand's thirteen calendar years to Japan's thirty-one, or twelve to twenty-nine years counting only the months in which the long–short portfolio is actually populated), depending on when a market's cross-section first supports the minimum training sample, so these magnitudes—below the contemporaneous U.S. alpha—should be read accordingly; the returns are gross of trading costs, and we make no implementability claim abroad. We note, however, that the absence of decay in the international alpha over its window is not anomalous: in a comprehensive study of 241 anomalies across 39 markets, Jacobs and Müller (2020) find that the post-publication decline pervasive in U.S. anomaly returns is essentially absent outside the United States.

6 Conclusion

Inspired by the Matrix Profile—a technique from the time series literature that identifies recurring motifs through exhaustive pairwise subsequence comparison—this paper develops a distance-based methodology to investigate whether price patterns in stock returns exhibit genuine recurrence and whether such recurrence generates predictable future returns. We extend the core principle of the Matrix Profile from its original single-series setting to the full cross-section of U.S. stocks from 1970 to 2025, comparing approximately 1.32 million 60-day return segments using GPU-accelerated computation. Unlike traditional technical analysis, which tests predefined chart patterns, our approach systematically searches for all recurring patterns without imposing prior beliefs about their shape or characteristics.

We introduce the Pattern Distance Ratio (PDR), which compares how closely a given return segment is matched in the real data versus a random walk benchmark. Normalizing by the random-walk benchmark is what turns a raw neighbor distance into a measure of recurrence: it removes the mechanical effects of a segment's generic shape commonness and of the changing density of the comparison set, so that PDR functions as a local density ratio comparable across shapes, stocks, and decades. Segments with PDR less than 1 exhibit stronger recurrence than would be expected by chance.

Our empirical analysis yields several key findings. First, we document strong persistence in pattern recurrence across time. Holding the comparison blocks fixed at one year, the correlation between in-sample and out-of-sample PDR is 0.308 (Pearson) at a one-year distance and decays only to 0.270 at a three-year distance, positive in at least 94% of block-pairs at every lag; lengthening the comparison blocks to three and five years raises the correlation to 0.486 and 0.603 through the precision gain of a larger comparison sample. In contrast, the random walk placebo test produces a correlation of 0.011—roughly thirty times smaller, and never above 0.017 across the twenty placebo configurations our five shuffled series permit—confirming that the methodology cannot manufacture persistence of the magnitude observed in real data. The persistence survives a substantially harder block-bootstrap null and a diversified neighbor rule, and it extends beyond the U.S.: the same construction yields positive persistence in twelve international equity markets.

Second, we find that segments with stronger recurrence demonstrate more persistent return predictability. Sign preservation analysis reveals that segments maintain directional predictions from in-sample to out-of-sample periods, with high-$|T|$ segments achieving preservation rates of 51.4–54.1% in the four most recurrent PDR quintiles, declining toward the null in the most novel quintile.

Third, we demonstrate that pattern-based predictability translates into economically significant portfolio returns. We map each return trajectory to its location in shape space—its Gaussian-kernel similarities to a grid of landmark trajectories—and predict future returns by ridge regression on this representation. The resulting long-short strategy earns large gross risk-adjusted alpha—roughly 6% to 9.5% per year across the CAPM and Fama-French three- through six-factor models and equal- and value-weighted portfolios—that remains significant in the most recent subperiods. The conditional evidence—its concentration in high-trading-cost stocks by scale-free measures of signal quality, and the post-2000 attenuation of the short-leg correction—is more consistent with a mispricing gradually competed away by arbitrage than with a priced risk, though the joint-hypothesis problem precludes a definitive verdict. Trading costs complete this picture: the strategy's high turnover leaves it a break-even one-way cost of about 15 to 21 basis points, at our estimates of the spreads it faced the net-of-cost alpha is negative, and the rolling gross return tracks the rolling cost of trading it over five decades—the mispricing persists inside the no-arbitrage band that transaction costs create, in the sense of Grossman and Stiglitz (1980) and Fama (1991). The predictability also survives the most demanding external-validity test we can construct: transplanted with every design choice frozen to twelve international markets, the strategy's raw equal-weighted long–short return is positive in all twelve, a country-average portfolio across those markets earns a significant alpha that is robust to restricting each market to its largest firms.

Our findings make several contributions to the asset pricing literature. First, we provide comprehensive evidence that stock price patterns exhibit genuine recurrence that persists across time and cannot be explained by temporally unstructured returns—nor, under the block-bootstrap surrogate, by stock-level distributional heterogeneity, volatility clustering, or factor comovement.

The contrast between real data and these benchmarks demonstrates that the recurrence reflects systematic market dynamics rather than statistical artifacts. Second, we establish a direct link between the strength of pattern recurrence (measured by PDR) and the persistence of return predictability. Patterns that recur more strongly exhibit more reliable future performance, suggesting that pattern recurrence captures an economically meaningful dimension of stock return dynamics. Third, we demonstrate that pattern-based predictability generates significant abnormal returns that remain unexplained by standard risk factors, including market, size, value, momentum, profitability, and investment factors. The orthogonality of our strategy to traditional factors indicates that price pattern recurrence represents a distinct source of return predictability.

The persistence of both pattern recurrence and return predictability across extended horizons indicates that the information embedded in price trajectories is not fully absorbed by standard pricing models—whether because it compensates for risks those models omit or because frictions bound its arbitrage. Our transaction-cost evidence favors the latter reading: the predictability is real, persistent, and statistically robust, but it lives within the band that trading frictions create, so its proper interpretation is as evidence about how prices form—and about the economic limits that allow structure to survive in them—rather than as a recipe for implementable arbitrage profits.

Several avenues for future research emerge from our analysis. First, investigating the behavioral or institutional mechanisms that generate persistent price patterns would enhance our understanding of why pattern recurrence predicts returns. Second, extending the analysis to other asset classes would test the generalizability of our findings beyond equities. Third, examining how pattern-based predictability varies with market conditions, liquidity, and information environments could provide insights into when and why patterns exhibit stronger recurrence. Finally, the broader structural properties of the trajectory space itself—the topology of trajectory clusters, the density dynamics of regions in shape space, and their interaction with market regimes—remain largely unexplored and constitute a natural research program of which this paper is the first step.

References

Abdi, Farshid and Angelo Ranaldo (2017) "A Simple Estimation of Bid-Ask Spreads from Daily Close, High, and Low Prices," *Review of Financial Studies*, 30 (12), 4437–4480.

Amihud, Yakov (2002) "Illiquidity and Stock Returns: Cross-Section and Time-Series Effects," *Journal of Financial Markets*, 5 (1), 31–56.

Ang, Andrew, Robert J Hodrick, Yuhang Xing, and Xiaoyan Zhang (2006) "The cross-section of volatility and expected returns," *The journal of finance*, 61 (1), 259–299.

Asness, Clifford S., Tobias J. Moskowitz, and Lasse Heje Pedersen (2013) "Value and Momentum Everywhere," *Journal of Finance*, 68 (3), 929–985.

Bajgrowicz, Pierre and Olivier Scaillet (2012) "Technical Trading Revisited: False Discoveries, Persistence Tests, and Transaction Costs," *Journal of Financial Economics*, 106 (3), 473–491, 10.1016/j.jfineco.2012.06.001.

Bali, Turan G., Nusret Cakici, and Robert F. Whitelaw (2011) "Maxing Out: Stocks as Lotteries and the Cross-Section of Expected Returns," *Journal of Financial Economics*, 99 (2), 427–446.

Banz, Rolf W (1981) "The relationship between return and market value of common stocks," *Journal of financial economics*, 9 (1), 3–18.

Bessembinder, Hendrik and Kalok Chan (1995) "The profitability of technical trading rules in the Asian stock markets," *Pacific-basin finance journal*, 3 (2-3), 257–284.

Brock, William, Josef Lakonishok, and Blake LeBaron (1992) "Simple technical trading rules and the stochastic properties of stock returns," *The Journal of finance*, 47 (5), 1731–1764.

Carhart, Mark M. (1997) "On Persistence in Mutual Fund Performance," *Journal of Finance*, 52 (1), 57–82.

Chaieb, Ines, Hugues Langlois, and Olivier Scaillet (2021) "Factors and Risk Premia in Individual International Stock Returns," *Journal of Financial Economics*, 141 (2), 669–692.

Chen, AJ, Gerard Hoberg, and Miao Ben Zhang (2026) "Haven't We Seen This Before? Return Predictions from 200 Years of News," Working paper.

Chen, Andrew Y. and Mihail Velikov (2023) "Zeroing In on the Expected Returns of Anomalies," *Journal of Financial and Quantitative Analysis*, 58 (3), 968–1004.

Chen, Andrew Y. and Tom Zimmermann (2022) "Open Source Cross-Sectional Asset Pricing," *Critical Finance Review*, 11 (2), 207–264.

Chen, Luyang, Markus Pelger, and Jason Zhu (2024) "Deep Learning in Asset Pricing," *Management Science*, 70 (2), 714–750.

Chui, Andy C. W., Sheridan Titman, and K. C. John Wei (2010) "Individualism and momentum around the world," *Journal of Finance*, 65 (1), 361–392.

Corwin, Shane A. and Paul Schultz (2012) "A Simple Way to Estimate Bid-Ask Spreads from Daily High and Low Prices," *Journal of Finance*, 67 (2), 719–760.

Cowles, Alfred (1933) "Can stock market forecasters forecast?" *Econometrica: Journal of the Econometric Society*, 309–324.

——— (1944) "Stock market forecasting," *Econometrica, Journal of the Econometric Society*, 206–214.

Da, Zhi, Qianqiu Liu, and Ernst Schaumburg (2014) "A closer look at the short-term return reversal," *Management Science*, 60 (3), 658–674.

De Bondt, Werner F. M. and Richard Thaler (1985) "Does the Stock Market Overreact?" *Journal of Finance*, 40 (3), 793–805.

Duffie, Darrell (2010) "Presidential address: Asset price dynamics with slow-moving capital," *The Journal of finance*, 65 (4), 1237–1267.

Fama, Eugene F (1970) "Efficient capital markets: A review of theory and empirical work," *The Journal of Finance*, 25 (2), 383–417.

Fama, Eugene F. (1991) "Efficient Capital Markets: II," *Journal of Finance*, 46 (5), 1575–1617.

Fama, Eugene F and Marshall E Blume (1966) "Filter rules and stock-market trading," *The Journal of Business*, 39 (1), 226–241.

Fama, Eugene F and Kenneth R French (1993) "Common risk factors in the returns on stocks and bonds," *Journal of financial economics*, 33 (1), 3–56.

Fama, Eugene F. and Kenneth R. French (2012) "Size, Value, and Momentum in International Stock Returns," *Journal of Financial Economics*, 105 (3), 457–472.

——— (2015) "A Five-Factor Asset Pricing Model," *Journal of Financial Economics*, 116 (1), 1–22.

——— (2017) "International Tests of a Five-Factor Asset Pricing Model," *Journal of Financial Economics*, 123 (3), 441–463.

Fama, Eugene F. and James D. MacBeth (1973) "Risk, Return, and Equilibrium: Empirical Tests," *Journal of Political Economy*, 81 (3), 607–636.

Frazzini, Andrea, Ronen Israel, and Tobias J. Moskowitz (2018) "Trading Costs," working paper, AQR Capital Management.

Gao, Xiaohui and Jay R. Ritter (2010) "The Marketing of Seasoned Equity Offerings," *Journal of Financial Economics*, 97 (1), 33–52.

George, Thomas J. and Chuan-Yang Hwang (2004) "The 52-Week High and Momentum Investing," *Journal of Finance*, 59 (5), 2145–2176.

Goulding, Christian L, Campbell R Harvey, and Michele G Mazzoleni (2024) "Breaking Bad Trends," *Financial Analysts Journal*, 80 (1), 84–98.

Green, Jeremiah, John R. M. Hand, and X. Frank Zhang (2017) "The Characteristics that Provide Independent Information about Average U.S. Monthly Stock Returns," *Review of Financial Studies*, 30 (12), 4389–4436.

Grossman, Sanford J. and Joseph E. Stiglitz (1980) "On the Impossibility of Informationally Efficient Markets," *American Economic Review*, 70 (3), 393–408.

Grundy, Bruce D. and J. Spencer Martin (2001) "Understanding the Nature of the Risks and the Source of the Rewards to Momentum Investing," *Review of Financial Studies*, 14 (1), 29–78.

Gu, Shihao, Bryan Kelly, and Dacheng Xiu (2020) "Empirical asset pricing via machine learning," *The Review of Financial Studies*, 33 (5), 2223–2273.

Han, Yufeng, Ke Yang, and Guofu Zhou (2013) "A new anomaly: The cross-sectional profitability of technical analysis," *Journal of Financial and Quantitative Analysis*, 48 (5), 1433–1461.

Han, Yufeng, Guofu Zhou, and Yingzi Zhu (2016) "A Trend Factor: Any Economic Gains from Using Information over Investment Horizons?" *Journal of Financial Economics*, 122 (2), 352–375.

Hou, Kewei, G. Andrew Karolyi, and Bong-Chan Kho (2011) "What Factors Drive Global Stock Returns?" *Review of Financial Studies*, 24 (8), 2527–2574.

Hou, Kewei and Tobias J. Moskowitz (2005) "Market Frictions, Price Delay, and the Cross-Section of Expected Returns," *Review of Financial Studies*, 18 (3), 981–1020.

Ince, Ozgur S. and R. Burt Porter (2006) "Individual equity return data from Thomson Datastream: Handle with care!," *Journal of Financial Research*, 29 (4), 463–479.

Jacobs, Heiko and Sebastian Müller (2020) "Anomalies across the globe: Once public, no longer existent?" *Journal of Financial Economics*, 135 (1), 213–230.

Jegadeesh, Narasimhan (1990) "Evidence of Predictable Behavior of Security Returns," *Journal of Finance*, 45 (3), 881–898.

Jegadeesh, Narasimhan and Sheridan Titman (1993) "Returns to buying winners and selling losers: Implications for stock market efficiency," *The Journal of finance*, 48 (1), 65–91.

Jiang, Jingwen, Bryan T Kelly, and Dacheng Xiu (2023) "(Re-)Imag(in)ing Price Trends," *The Journal of Finance*, 78 (6), 3193–3249.

Keim, Donald B (1983) "Size-related anomalies and stock return seasonality: Further empirical evidence," *Journal of financial economics*, 12 (1), 13–32.

Kelly, Bryan, Semyon Malamud, and Kangying Zhou (2024) "The virtue of complexity in return prediction," *The Journal of Finance*, 79 (1), 459–503.

Kwon, Ki-Yeol and Richard J Kish (2002) "Technical trading strategies and return predictability: NYSE," *Applied Financial Economics*, 12 (9), 639–653.

Kyle, Albert S. and Anna A. Obizhaeva (2016) "Market Microstructure Invariance: Empirical Hypotheses," *Econometrica*, 84 (4), 1345–1404.

Lehmann, Bruce N. (1990) "Fads, Martingales, and Market Efficiency," *Quarterly Journal of Economics*, 105 (1), 1–28.

Leippold, Markus, Qian Wang, and Wenyu Zhou (2022) "Machine learning in the Chinese stock market," *Journal of Financial Economics*, 145 (2), 64–82.

Lin, Jessica, Eamonn Keogh, Stefano Lonardi, and Pranav Patel (2002) "Finding motifs in time series," in *Proceedings of the 2nd Workshop on Temporal Data Mining*, 53–68.

Lo, Andrew W, Harry Mamaysky, and Jiang Wang (2000) "Foundations of technical analysis: Computational algorithms, statistical inference, and empirical implementation," *The journal of finance*, 55 (4), 1705–1765.

Loftsgaarden, Don O. and Charles P. Quesenberry (1965) "A Nonparametric Estimate of a Multivariate Density Function," *The Annals of Mathematical Statistics*, 36 (3), 1049–1051.

McLean, R David and Jeffrey Pontiff (2016) "Does academic research destroy stock return predictability?" *The Journal of Finance*, 71 (1), 5–32.

McLean, R. David, Jeffrey Pontiff, and Akiko Watanabe (2009) "Share issuance and cross-sectional returns: International evidence," *Journal of Financial Economics*, 94 (1), 1–17.

Medhat, Mamdouh and Maik Schmeling (2022) "Short-term Momentum," *Review of Financial Studies*, 35 (3), 1480–1526.

Menkhoff, Lukas (2010) "The use of technical analysis by fund managers: International evidence," *Journal of Banking & Finance*, 34 (11), 2573–2586.

Nagel, Stefan (2025) "Seemingly Virtuous Complexity in Return Prediction," Working Paper 34104, National Bureau of Economic Research.

Neely, Christopher J, David E Rapach, Jun Tu, and Guofu Zhou (2014) "Forecasting the equity risk premium: the role of technical indicators," *Management science*, 60 (7), 1772–1791.

Newey, Whitney K. and Kenneth D. West (1987) "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix," *Econometrica*, 55 (3), 703–708.

Novy-Marx, Robert and Mihail Velikov (2016) "A Taxonomy of Anomalies and Their Trading Costs," *Review of Financial Studies*, 29 (1), 104–147.

Odean, Terrance (1998) "Volume, volatility, price, and profit when all traders are above average," *The journal of finance*, 53 (6), 1887–1934.

Politis, Dimitris N. and Joseph P. Romano (1994) "The stationary bootstrap," *Journal of the American Statistical Association*, 89 (428), 1303–1313.

Rahimi, Ali and Benjamin Recht (2008) "Random features for large-scale kernel machines," in *Advances in Neural Information Processing Systems*, 20, 1177–1184.

Shefrin, Hersh and Meir Statman (1985) "The disposition to sell winners too early and ride losers too long: Theory and evidence," *The Journal of finance*, 40 (3), 777–790.

Shleifer, Andrei and Robert W Vishny (1997) "The limits of arbitrage," *The Journal of Finance*, 52 (1), 35–55.

Stambaugh, Robert F, Jianfeng Yu, and Yu Yuan (2012) "The short of it: Investor sentiment and anomalies," *Journal of financial economics*, 104 (2), 288–302.

——— (2015) "Arbitrage asymmetry and the idiosyncratic volatility puzzle," *The Journal of Finance*, 70 (5), 1903–1948.

Sullivan, Ryan, Allan Timmermann, and Halbert White (1999) "Data-Snooping, Technical Trading Rule Performance, and the Bootstrap," *The Journal of Finance*, 54 (5), 1647–1691, 10.1111/0022-1082.00163.

Watanabe, Akiko, Yan Xu, Tong Yao, and Tong Yu (2013) "The asset growth effect: Insights from international equity markets," *Journal of Financial Economics*, 108 (2), 529–563.

Williams, Christopher and Matthias Seeger (2001) "Using the Nyström method to speed up kernel machines," in *Advances in Neural Information Processing Systems*, 13, 682–688.

Yeh, Chin-Chia Michael, Yan Zhu, Liudmila Ulanova et al. (2016) "Matrix profile I: All pairs similarity joins for time series: A unifying view that includes motifs, discords and shapelets," in *2016 IEEE 16th International Conference on Data Mining (ICDM)*, 1317–1322, IEEE.

Zhu, Yingzi and Guofu Zhou (2009) "Technical analysis: An asset allocation perspective on the use of moving averages," *Journal of financial economics*, 92 (3), 519–544.

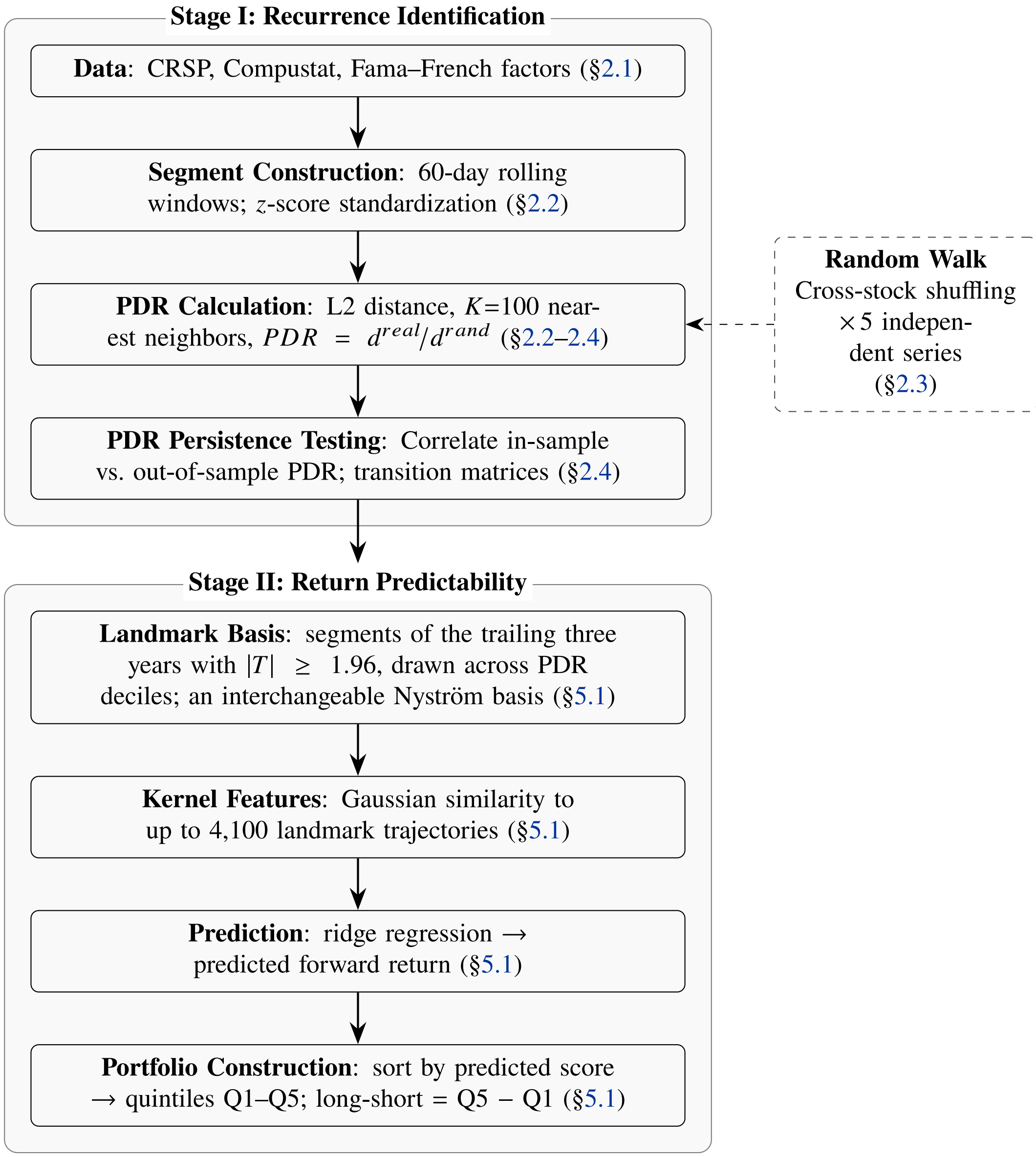

Figure 1: Overview of the methodology. The procedure consists of two stages. *Stage I (Recurrence Identification)* constructs return segments, computes the Pattern Distance Ratio (PDR) relative to a random walk benchmark, and tests whether recurrence persists over time. *Stage II (Return Predictability)* fixes a basis of landmark trajectories (drawn from the Stage I recurrence grid, although any basis covering the data performs equivalently; Section 5.1), represents each trajectory by its kernel similarities to the landmarks, and constructs portfolios from ridge predictions. The dashed box denotes the random walk benchmark (Section 2.3). Section references are shown in parentheses.

Figure 2: PDR Persistence Comparison

This figure presents the persistence of the Pattern Distance Ratio (PDR) across holding horizons. The figure is arranged as a two-row grid in which each column corresponds to one horizon: the 1-Year, 2-Year, 3-Year, and 5-Year block lengths and a Random Walk placebo test. In the top row, each panel is a scatter plot of out-of-sample PDR against in-sample PDR, with one point per return segment (a random subsample of 3,000 segments is plotted for legibility); the dashed 45-degree line denotes perfect persistence, the dotted gray lines mark $PDR = 1$, and the pooled Pearson correlation r is reported in each panel. In the bottom row, each panel sorts segments into deciles of in-sample PDR and plots the mean in-sample PDR (circles) and the mean out-of-sample PDR (squares) within each decile, with the shaded band indicating the gap between the two. For the real horizons, points cluster along the 45-degree line and the in-sample and out-of-sample decile curves track each other closely, with the correlation rising as the comparison block lengthens (Panel D of Table 1 separates this comparison-sample effect from the effect of temporal distance); under the Random Walk benchmark the scatter is uncorrelated and the out-of-sample decile curve is flat, as expected (the in-sample curve rises mechanically, since deciles are sorted on it). All analyses use data from 1971–2025.

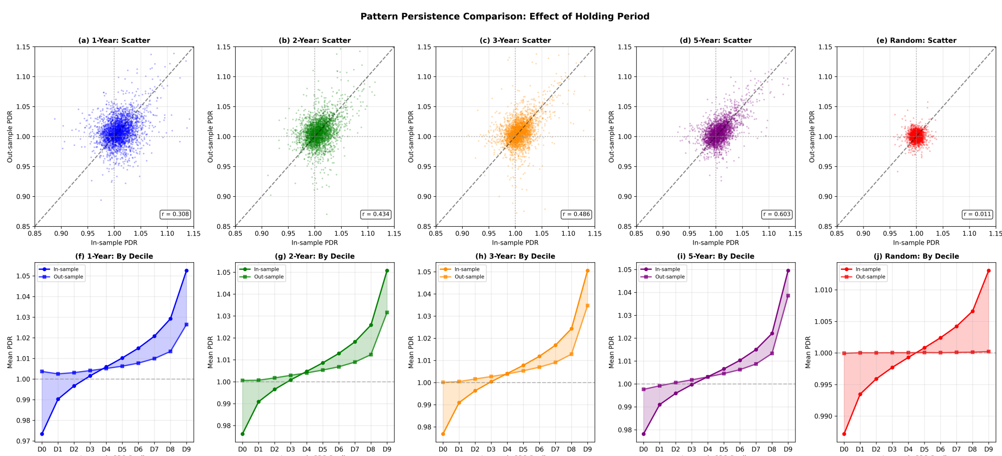

Figure 3: Yearly Correlation Trends

This figure plots the Pearson correlation between in-sample and out-of-sample Pattern Distance Ratio (PDR) for each year from 1970 to 2023, with gray shaded areas indicating NBER recession periods and the horizontal dotted line marking the full-sample mean of 0.317. The series shows notable temporal variation, with particularly strong persistence during the 1990s and weaker persistence in the 2000s and 2010s. Comparing recession periods (N=14 years) to expansion periods (N=40 years), the gap in the yearly correlation (0.272 versus 0.333) is well under half a year-to-year standard deviation; with fourteen recession years the comparison has little power. This suggests that pattern persistence is robust across economic cycles rather than driven by regime switching. The directional counterpart of this persistence—sign preservation of the neighbor-outcome statistic—is reported by PDR quintile in Table 3.

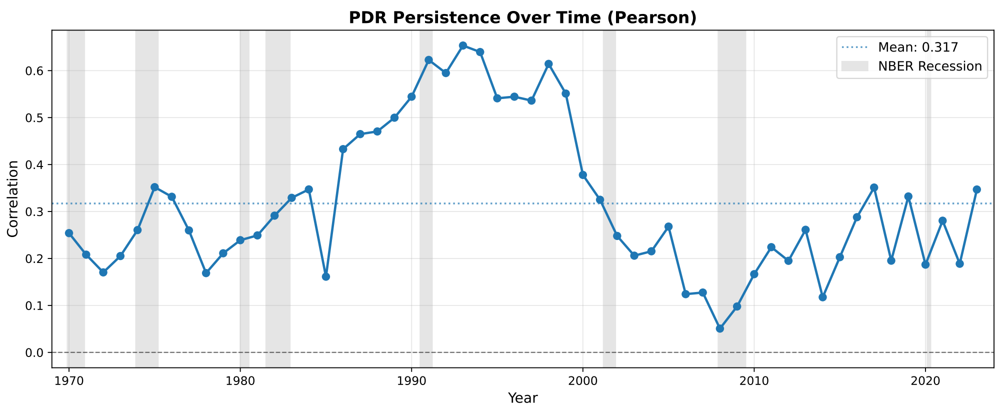

Table 1: PDR Persistence Comparison

This table presents the persistence analysis of Pattern Distance Ratio (PDR) compared to a Random Walk placebo test. In the 1-Year, 2-Year, 3-Year, and 5-Year specifications the sample is partitioned into non-overlapping one-, two-, three-, and five-year blocks, respectively (54, 27, 18, and 10 consecutive block-pairs; the 5-Year sample is smaller because its final block, 2021–2025, has no successor block, whereas the other specifications end in a one-year 2025 block), and each block's in-sample PDR is correlated with the same segments' out-of-sample PDR measured against the next block; note that lengthening the block moves *two* things at once—the temporal distance between the compared blocks and the size of the comparison sample from which nearest neighbors are drawn. Panel A reports the mean PDR by in-sample quintile under each specification (Q1 = most recurrent, Q5 = most novel). Panel B shows overall correlation statistics between in-sample and out-of-sample PDR. Panel C reports the mean of the correlations computed block-pair by block-pair. Panel D isolates the effect of temporal distance alone: the comparison block is held fixed at one year and only the horizon h is varied, correlating the in-sample PDR of annual block $\mathcal{D}_y$ with the out-of-sample PDR measured against $\mathcal{D}_{y+h}$ for $h = 1, 2, 3, 5$. The Random Walk column is a single realization of a shuffled benchmark; because its persistence is itself a random variable, the informative comparison with the real-data columns is one of magnitude—placebo mean correlations range from −0.003 to +0.017 across the twenty configurations our five independent shuffled series permit, against 0.308 in real data (Section 3.1). All analyses use data from 1971–2025.

Panel A: Mean PDR by In-Sample Quintile					
Quintile	1-Year Blocks	2-Year Blocks	3-Year Blocks	5-Year Blocks	Random Walk
Q1	0.982	0.984	0.984	0.984	0.990
Q2	1.000	0.999	0.999	0.998	0.997
Q3	1.009	1.007	1.006	1.005	1.000
Q4	1.019	1.016	1.014	1.012	1.003
Q5	1.038	1.036	1.035	1.034	1.010
All segments	1.010	1.008	1.008	1.007	1.000
Panel B: Overall Correlation—Pooled					
Metric	1-Year Blocks	2-Year Blocks	3-Year Blocks	5-Year Blocks	Random Walk
Pooled Pearson r	0.308	0.434	0.486	0.603	0.011
Pooled Spearman r	0.251	0.323	0.356	0.444	0.008
N samples	1,295,590	1,295,590	1,295,590	1,200,545	1,295,608
Panel C: Block-Pair Consistency					
Metric	1-Year Blocks	2-Year Blocks	3-Year Blocks	5-Year Blocks	Random Walk
Mean Spearman	0.233	0.323	0.361	0.450	0.010
Panel D: Fixed Annual Blocks, Varying Horizon					
Horizon h	1 year	2 years	3 years	5 years	
Pooled Pearson r	0.308	0.275	0.270	0.257	
Pooled Spearman r	0.251	0.220	0.216	0.193	
Mean Spearman	0.233	0.218	0.207	0.198	
N samples	1,295,590	1,272,410	1,248,355	1,200,545	

Table 2: PDR Quintile Transition Matrices

This table presents transition matrices showing the probability that a segment in quintile q_1 (in-sample) moves to quintile q_2 (out-of-sample). Rows represent in-sample quintiles, columns represent out-of-sample quintiles. Higher diagonal values indicate stronger persistence. Panel A shows 1-year transitions, Panel B shows 2-year transitions, Panel C shows 3-year transitions, and Panel D shows Random Walk placebo. All analyses use data from 1971–2025.

Panel A: 1-Year

	Q1	Q2	Q3	Q4	Q5
Q1	0.269	0.234	0.196	0.168	0.133
Q2	0.229	0.248	0.227	0.183	0.112
Q3	0.196	0.222	0.229	0.212	0.140
Q4	0.172	0.181	0.208	0.232	0.206
Q5	0.134	0.113	0.139	0.204	0.408

Panel B: 2-Year

	Q1	Q2	Q3	Q4	Q5
Q1	0.296	0.248	0.197	0.157	0.102
Q2	0.233	0.257	0.230	0.182	0.099
Q3	0.191	0.223	0.235	0.218	0.134
Q4	0.162	0.174	0.211	0.245	0.207
Q5	0.118	0.099	0.127	0.199	0.458

Panel C: 3-Year

	Q1	Q2	Q3	Q4	Q5
Q1	0.310	0.251	0.193	0.152	0.094
Q2	0.238	0.259	0.232	0.179	0.092
Q3	0.189	0.227	0.240	0.219	0.125
Q4	0.155	0.173	0.214	0.254	0.204
Q5	0.108	0.091	0.121	0.196	0.485

Panel D: Random Walk

	Q1	Q2	Q3	Q4	Q5
Q1	0.226	0.186	0.183	0.185	0.220
Q2	0.187	0.209	0.210	0.208	0.186
Q3	0.183	0.211	0.213	0.209	0.183
Q4	0.186	0.207	0.211	0.210	0.186
Q5	0.217	0.186	0.183	0.188	0.226

Table 3: Sign Preservation Rate by PDR Quintile and Statistical Significance

This table presents sign preservation rates across Pattern Distance Ratio (PDR) quintiles for different holding periods. Panel A reports results for segments with high neighbor-outcome scores ($|T| \geq 1.96$), and Panel B for segments with low scores ($|T| < 1.96$); see Appendix B for the interpretation of the cutoff. Sign preservation measures the proportion of segments for which the sign of the T-statistic is the same in-sample and out-of-sample. Q1 represents segments with the lowest PDR (highest recurrence), while Q5 represents segments with the highest PDR (lowest recurrence). The "Adj. Null" row reports the independence-adjusted null benchmark, computed as $P(\text{in}>0) \times P(\text{out}>0) + P(\text{in}<0) \times P(\text{out}<0)$, where $P(\text{out}>0)$ is the unconditional probability from the full sample. Block lengths: one-, two-, three-, and five-year blocks, as in Table 1, 1971–2025.

Panel A: High Statistical Significance ($|T| \geq 1.96$)

Quintile	1-Year	2-Year	3-Year	5-Year
Q1 (Low PDR)	53.07%	53.44%	52.50%	52.15%
Q2	53.56%	54.13%	53.21%	53.34%
Q3	53.08%	53.19%	52.73%	52.47%
Q4	52.84%	51.79%	52.40%	51.37%
Q5 (High PDR)	50.63%	51.21%	50.79%	50.51%
Adj. Null	50.24%	50.29%	50.39%	50.38%

Panel B: Low Statistical Significance ($|T| < 1.96$)

Quintile	1-Year	2-Year	3-Year	5-Year
Q1 (Low PDR)	51.13%	51.16%	51.00%	51.06%
Q2	51.11%	51.26%	51.13%	50.74%
Q3	51.04%	50.86%	50.77%	50.53%
Q4	50.92%	50.56%	50.56%	50.49%
Q5 (High PDR)	50.46%	49.83%	50.07%	50.23%
Adj. Null	50.02%	50.02%	50.02%	50.02%

Table 4: Long–Short Alphas: Factor Models and Subsample Stability

Annualized alphas (%) of the long–short portfolio ($Q5 - Q1$), equal- and value-weighted (winsorized market equity), over the out-of-sample period 1975–2025. *Panel A* reports alphas from the CAPM, the Fama–French three- and five-factor models (Fama and French, 1993, 2015), the Carhart four-factor model (Carhart, 1997), the five-factor model augmented with momentum (FF6), FF6 augmented with the Fama–French short-term *monthly* reversal factor (ST-Rev; Jegadeesh 1990; Lehmann 1990), taken from Kenneth French's data library, and that model further augmented with a *weekly* reversal factor (WST-Rev). French publishes no weekly reversal factor, so WST-Rev is built inside our own universe under the strategy's own rule: each month segments are sorted on the compound return over the last five days of the formation window, the bottom quintile is bought and the top quintile sold, and the position is held for twenty days; the equal-weighted factor is used in the equal-weighted columns and the value-weighted factor in the value-weighted columns. Following Da et al. (2014), the estimation target is residualized on the segment's own five-day and 21-day reversal characteristics and the fitted signal is orthogonalized to the same two characteristics before portfolios are formed, so the two reversal factors serve as a check rather than a correction. *Panels B and C* repeat the same seven models over the 2000–2025 and 2009–2025 subsamples. Section 5.2 describes the signal, the weekly horizon, and the factor loadings; firm-size heterogeneity is in Table 5; the long/short decomposition is reported in Section 5.4. t-statistics in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Panel A: Factor models, full sample (1975–2025)		
Model	Equal-weighted	Value-weighted
CAPM	9.57*** (10.96)	6.84*** (5.76)
FF3	9.44*** (10.80)	6.69*** (5.63)
FF4	8.86*** (10.07)	6.43*** (5.32)
FF5	8.79*** (9.86)	6.07*** (4.99)
FF6	8.39*** (9.38)	5.92*** (4.81)
FF6 + ST-Rev	8.54*** (9.52)	5.81*** (4.70)
FF6 + ST-Rev + WST-Rev	7.30*** (7.69)	6.33*** (4.96)
Panel B: Factor models, 2000–2025		
Model	Equal-weighted	Value-weighted
CAPM	10.25*** (7.01)	9.17*** (4.78)
FF3	10.02*** (6.90)	8.87*** (4.67)
FF4	9.63*** (6.65)	8.77*** (4.59)
FF5	9.00*** (6.09)	8.05*** (4.13)
FF6	8.78*** (5.96)	8.03*** (4.09)
FF6 + ST-Rev	8.81*** (5.99)	8.00*** (4.09)
FF6 + ST-Rev + WST-Rev	8.19*** (5.50)	8.51*** (4.33)
Panel C: Factor models, 2009–2025		
Model	Equal-weighted	Value-weighted
CAPM	9.00*** (5.34)	8.74*** (4.35)
FF3	8.98*** (5.27)	8.65*** (4.27)
FF4	8.73*** (5.19)	8.32*** (4.17)
FF5	8.66*** (5.05)	8.42*** (4.11)
FF6	8.45*** (4.99)	8.12*** (4.04)
FF6 + ST-Rev	8.29*** (4.88)	7.87*** (3.92)
FF6 + ST-Rev + WST-Rev	8.43*** (4.96)	8.29*** (4.17)

Table 5: Signal Quality by Firm Size

This table reports the accuracy of the signal within firm-size terciles (NYSE breakpoints: the 33rd and 67th percentiles of NYSE market equity each month, applied to all stocks). For the long–short portfolio formed inside each tercile it reports the monthly cross-sectional rank information coefficient (rank-IC), computed as the unweighted Spearman correlation between the signal and the subsequent 20-day return over all stocks in the tercile, and annualized Sharpe ratios under both weighting schemes. Because a raw long–short alpha is mechanically larger where returns are more dispersed, a size gradient in the alpha is not by itself evidence of a gradient in signal quality; these scale-free measures are reported for that reason. Results are reported over the full out-of-sample period and the 2000–2025 and 2009–2025 subsamples. t-statistics in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels.

	Small	Mid	Large
Full (1975–2025)			
rank-IC	0.028*** (12.6)	0.023*** (7.7)	0.021*** (5.1)
Sharpe (EW/VW)	1.51/1.32	0.99/0.96	0.67/0.57
2000–2025			
rank-IC	0.025*** (6.6)	0.025*** (5.3)	0.023*** (3.6)
Sharpe (EW/VW)	1.25/1.08	0.89/0.88	0.65/0.73
2009–2025			
rank-IC	0.022*** (4.7)	0.023*** (4.0)	0.028*** (3.6)
Sharpe (EW/VW)	1.07/0.83	0.99/0.98	0.70/0.89

Table 6: Is the Signal Spanned by Technical Rules and Standard Predictors?

This table reports monthly Fama–MacBeth cross-sectional regressions of the 20-day holding return (in percent) on the engine's predicted-return signal and competing predictors, in the layout of Han et al. (2016), Table 12, with the control set of Jiang et al. (2023), Table VII (we use idiosyncratic rather than total volatility) plus the maximum daily return, long-term reversal, book-to-market, and share turnover. Column (1) adds the technical rules to the signal, column (2) the standard predictors as well, column (3) the weekly reversal characteristic, and column (4) the trend forecast of Han et al. (2016) and the price delay of Hou and Moskowitz (2005), both from the open-source panel of Chen and Zimmermann (2022). The signal alone carries a slope of 0.238 (t = 8.30); two further specifications are available on request. *Signal* is the within-month percentile rank of the engine's score, orthogonalized to the five- and 21-day reversal characteristics before ranking (Section 5.2). The technical rules are computed on the return-implied price path of the same 60-day window the engine observes—*MA ratio* is the last price over the 60-day average price and *MA crossover* the 5-day over the 50-day average (Brock et al., 1992; Han et al., 2013)—so a mid-window split cannot distort them. The standard predictors are 21-day reversal, 12–2 momentum, long-term reversal over months −60 to −25 (De Bondt and Thaler, 1985), IVOL (Ang et al., 2006), the largest daily return of the last 21 days (Bali et al., 2011), market beta, log market equity, log book-to-market, the 52-week-high ratio (George and Hwang, 2004), and five liquidity measures averaged over the prior twelve months: Amihud (2002) illiquidity, share turnover, the effective spread that Table 7 charges trades at (Corwin and Schultz, 2012; Abdi and Ranaldo, 2017), log dollar volume, and the fraction of zero-volume days. *Reversal (weekly, 5-day)* is the compound return over the last five days of the window, the horizon of the WST-Rev factor of Table 4. Every regressor is winsorized at the 1st and 99th percentiles within the month and then standardized (Green et al., 2017; Medhat and Schmeling, 2022), so a slope is the change in the 20-day return, in percent, per one-standard-deviation move. The regressions are equal-weighted; Table A7 is the value-weighted (WLS, market-equity weights) counterpart. Columns (1)–(3) share a common sample: 1975–2025, 611 months, 1,673 stocks per month. The open-source panel ends in December 2024, so column (4) uses 595 months and 1,610 stocks per month. Newey–West (12 lags) t-statistics in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Regressor	(1)	(2)	(3)	(4)
Signal (engine prediction)	0.193*** (7.09)	0.152*** (8.21)	0.182*** (8.33)	0.155*** (7.09)
MA ratio	−1.076*** (−8.06)	−1.011*** (−8.62)	−0.408*** (−4.57)	−0.334*** (−3.58)
MA crossover	0.897*** (7.22)	0.895*** (8.66)	0.321*** (3.80)	0.220** (2.29)
Reversal (monthly, 21-day)		−0.275*** (−5.53)	−0.160*** (−3.42)	−0.108** (−2.24)
Momentum (12–2)		0.243*** (3.45)	0.232*** (3.35)	0.268*** (5.12)
Long-term reversal (60–25)		−0.042 (−1.38)	−0.042 (−1.42)	−0.046 (−1.59)
IVOL		−0.240*** (−4.46)	−0.242*** (−4.58)	−0.288*** (−5.35)
Max		0.026 (0.92)	0.034 (1.24)	0.065*** (2.81)
Beta		0.062 (0.95)	0.065 (1.06)	0.061 (0.91)
log market equity		−0.342*** (−3.34)	−0.361*** (−3.53)	−0.288*** (−3.32)
log book-to-market		0.026 (0.58)	0.021 (0.47)	0.014 (0.32)
Illiquidity (Amihud)		0.069*** (2.87)	0.073*** (3.01)	0.087*** (3.96)
Turnover		−0.052 (−0.96)	−0.060 (−1.15)	−0.059 (−1.14)
Bid–ask spread		0.049 (1.05)	0.036 (0.79)	0.024 (0.52)
Dollar volume (log)		0.213** (2.17)	0.233** (2.39)	0.178** (2.19)
Zero-volume days		−0.017 (−1.55)	−0.017 (−1.53)	−0.028** (−2.30)
Trend forecast (HZZ)				0.077* (1.75)
Price delay				0.001 (0.05)
52-week high		0.044 (0.88)	0.019 (0.40)	−0.052 (−1.15)
Reversal (weekly, 5-day)			−0.298*** (−6.72)	−0.258*** (−6.47)
Avg. adj. R^2 (%)	2.44	10.05	10.28	10.86

Table 7: Implementability: Turnover and Break-Even Trading Costs

This table reports how much the long–short strategy trades and the trading cost at which its alpha would be exhausted, in the design of Grundy and Martin (2001), Table 6. *Monthly rebalancing* is the baseline strategy of Table 4. Panel B repeats it inside the largest of the NYSE size terciles of Table 5, so that both legs hold only large firms. *Gross* is the annualized FF6 alpha (%) with its t-statistic. *Turnover* is one-way turnover per leg in percent per month, computed after drifting the previous month's weights by their realized holding-period returns, so that passive appreciation is not counted as trading. The two *break-even* columns report the uniform one-way cost per dollar traded, in basis points, at which the strategy's profit would be exhausted, under the two conventions of Grundy and Martin (2001): the cost at which the net FF6 alpha is exactly zero, which is the gross alpha divided by the dollars traded per month—every purchase and sale, on both legs—per dollar held on a side; and the cost at which the net alpha's t-statistic falls to 1.96, so that the profit can no longer be distinguished from zero. The second is always the lower of the two, since significance is lost before the profit is. Both use only the alpha, its standard error, and the turnover, not any estimate of the spreads the strategy actually paid, and are to be compared with whatever execution cost a reader considers realistic; the round-trip equivalent is twice the figure shown. Section 5.5 compares these hurdles with our own estimates of the effective spreads and with the price-impact capacity of the strategy. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Panel A: Monthly rebalancing, all stocks

Sample		Gross FF6 α (%/yr)	Turnover (%/mo)	Break-even one-way cost (bp)	
				$\alpha = 0$	$t = 1.96$
Full (1975–2025)	EW	8.39*** (9.38)	82	21.3	16.7
	VW	5.92*** (4.81)	84	14.6	8.6
2000–2025	EW	8.78*** (5.96)	82	22.3	15.0
	VW	8.03*** (4.09)	84	19.9	10.4
2009–2025	EW	8.45*** (4.99)	82	21.4	13.0
	VW	8.12*** (4.04)	84	20.2	10.4

Panel B: Monthly rebalancing, largest NYSE size tercile

Sample		Gross FF6 α (%/yr)	Turnover (%/mo)	$\alpha = 0$	$t = 1.96$
Full (1975–2025)	EW	5.01*** (4.04)	82	12.7	6.5
	VW	4.43*** (3.40)	83	11.2	4.7
2000–2025	EW	5.97*** (2.99)	82	15.2	5.3
	VW	7.01*** (3.41)	82	17.7	7.5
2009–2025	EW	5.08** (2.38)	82	12.9	2.3
	VW	6.88*** (3.38)	82	17.4	7.3

Table 8: The Engine in International Markets: One-Shot Out-of-Sample Deployment

The engine of Section 5.6, with every design choice frozen at its U.S. value, deployed in the twelve international markets of Table A5. *Panel A* reports **annualized** long–short ($Q5 - Q1$) alphas (%), equal- and value-weighted, for the full screened universe. Each market is benchmarked against its own regional five-factor-plus-momentum model (Fama and French, 2017); markets are grouped into French's Developed and Emerging universes and ordered by Stage I PDR persistence within each. *Global* is the country-average portfolio of Chui et al. (2010), reported as a raw return; *Developed* and *Emerging* split it into the six markets in French's Developed universe and the six in his Emerging universe, each reported as an alpha against its own regional factor model. t-statistics are ordinary least squares. *Panel B* reports pooled cross-country Fama–MacBeth regressions of the 20-day forward return (demeaned within country-month) on within-country-month percentile ranks of the signal and controls, with Newey–West t-statistics on 335 monthly slopes. These are coefficients per holding period, not annualized alphas. Where Compustat Global book equity is missing or negative (24% of firm-months) we set book-to-market to zero and add an availability indicator, following McLean et al. (2009). The control set is that of the cross-country stock-level design of McLean et al. (2009); the last column adds Amihud (2002) illiquidity, computed from the same daily files as the returns, so that the result cannot be read as a liquidity premium. Section 5.6 gives the construction of the global series, the control set, the out-of-sample windows, and the short-horizon reversal column. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Panel A: Long–short returns by market

Market	Months	Full universe EW	Full universe VW
Developed markets			
Hong Kong	188	7.46*** (3.62)	9.31*** (2.88)
Japan	342	4.60*** (3.95)	5.24*** (2.91)
France	268	3.01** (1.97)	3.30 (1.10)
Australia	244	10.02*** (4.00)	1.76 (0.57)
Germany	179	2.98 (1.23)	−2.73 (−0.71)
Sweden	162	9.49*** (3.32)	3.70 (1.02)
Emerging markets			
Thailand	156	6.23*** (2.96)	12.32*** (2.73)
Malaysia	261	9.93*** (4.83)	12.70*** (4.79)
South Korea	273	3.79** (2.42)	−0.03 (−0.01)
India	140	11.42*** (3.97)	2.78 (0.69)
Taiwan	257	2.96** (2.02)	2.09 (0.85)
China	233	3.24 (1.63)	2.74 (0.83)
Global, raw return	264	6.34*** (10.82)	4.97*** (5.66)
Developed, regional-factor alpha	249	6.82*** (7.66)	4.78*** (3.48)
Emerging, regional-factor alpha	264	5.96*** (7.11)	5.13*** (3.99)

Panel B: Pooled cross-country Fama–MacBeth regressions

	Univariate	Multivariate	+ Weekly rev.	+ Illiquidity
Signal (engine prediction)	0.53*** (7.38)	0.53*** (7.83)	0.55*** (8.01)	0.56*** (8.42)
log market equity		−0.98*** (−4.11)	−0.93*** (−3.95)	−0.48 (−1.12)
Book-to-market		0.68*** (3.12)	0.68*** (3.15)	0.63*** (2.98)
B/M available		0.41*** (2.96)	0.41*** (2.95)	0.39*** (2.82)
Reversal (monthly)		−0.51** (−2.43)	−0.08 (−0.42)	−0.05 (−0.28)
Reversal (weekly)			−1.07*** (−6.25)	−1.09*** (−6.83)
Momentum (12–2)		0.46 (1.57)	0.51* (1.77)	0.56* (1.96)
Illiquidity (Amihud)				0.59* (1.82)

Appendix

A. Summary of Variables and Notations

Table A1: Summary of Variables and Notations

Symbol	Description
Data and Sample Construction	
$r_{i,t}$	Daily return of stock i on day t
$P_{j,t}$	Closing price of stock j on day t
$SHROUT_{j,t}$	Shares outstanding of stock j on day t
$ME_{j,t}$	Market equity: $\lvert P_{j,t}\rvert \times SHROUT_{j,t}$
BE	Book equity: $SEQ + TXDITC - PS$
BM	Book-to-market ratio: $BE_{y-1}/ME_{Dec,y-1}$
$p(j)$	FF25 portfolio to which stock j is assigned
Return Segment Construction and Standardization	
$\mathbf{R}_{i,t}$	60-day return segment for stock i ending at date t: $(r_{i,t-59}, \ldots, r_{i,t})$
$\bar{R}_n$	Within-window mean: $(1/60)\sum_{s=t-59}^{t} r_{i,s}$
σ_n	Within-window standard deviation
$\tilde{R}_{n,k}$	Standardized return: $(R_{n,k} - \bar{R}_n)/\sigma_n$
Distance and Nearest Neighbors	
$d_{m,n}$	Z-normalized Euclidean distance between standardized segments m and n (Equation 2)
K	Number of nearest neighbors ($K = 100$)
ν_k^{real}	k-th nearest real-data neighbor of segment n
$\nu_k^{\text{rand,m}}$	k-th nearest neighbor from random walk series m
$\mathcal{D}_y$	Set of all segments whose formation window ends in year y
Random Walk Benchmark	
$\mathcal{R}_y$	Pool of all daily returns across all stocks in year y
N_y	Total number of return observations in year y: $\lvert\mathcal{R}_y\rvert$
Pattern Distance Ratio (PDR)	
$\bar{d}_n^{\text{real}\to\text{real}}$	Mean distance from segment n to its K nearest real neighbors
$\bar{d}_n^{\text{real}\to\text{rand}}$	Mean distance to nearest neighbors averaged across 3 random series
PDR_n	Pattern Distance Ratio: $\bar{d}_n^{\text{real}\to\text{real}}/\bar{d}_n^{\text{real}\to\text{rand}}$
Return Measures and Predictability	
R_n^{FF25}	FF25-adjusted cumulative 20-day return for segment n
R_n^{raw}	Unadjusted cumulative 20-day return
$r_{p(j),t+s}$	Daily return of the FF25 benchmark portfolio for stock j on day $t+s$
T_n	T-statistic of segment n's neighbors' future FF25-adjusted returns
$\bar{R}_n^{\text{FF25}}$	Mean FF25-adjusted return of segment n's K nearest neighbors

Continued on next page

Table A1 – continued from previous page

Symbol	Description
$\hat{\sigma}_n^{\text{FF25}}$	Standard deviation of neighbors' FF25-adjusted returns
Persistence Testing	
PDR_n^{in}	In-sample PDR (from $\mathcal{D}_y$ vs. $\mathcal{D}_y$)
$PDR_n^{\text{out,h}}$	Out-of-sample PDR at horizon h (from $\mathcal{D}_y$ vs. $\mathcal{D}_{y+h}$)
$T_n^{\text{in}}, T_n^{\text{out,h}}$	In-sample and out-of-sample T-statistics
$\mathbf{T}^h$	PDR quintile transition matrix at horizon h
T_{q_1,q_2}^h	Transition probability from quintile q_1 to q_2
SP^h	Sign preservation rate at horizon h
PDR_n^{placebo}	Placebo PDR: $\bar{d}_n^{\text{rand},5\to\text{rand},5} / \bar{d}_n^{\text{rand},5\to\text{rand}}$
Portfolio Construction (Appendix B)	
$\mathcal{S}_Y$	Landmark set for test year Y: 410 segments drawn from each PDR decile of the pool of years $Y-3$ to $Y-1$ with $\lvert T\rvert \geq 1.96$ (the recurrence grid)
$R_{q,\tau}^{\text{EW}}, R_{q,\tau}^{\text{VW}}$	Equal-weighted and value-weighted portfolio returns for quintile q
$R_{LS,\tau}$	Long-short portfolio return (Q5 − Q1)
Factor Models	
$MktRf_\tau$	Market excess return
SMB_τ, HML_τ	Size and value factors
UMD_τ	Momentum factor
RMW_τ, CMA_τ	Profitability and investment factors

B. Portfolio Construction Methodology

This appendix provides the formal definitions of the return measures, the segment-selection filters that define the recurrence grid, and the portfolio-formation and factor-model conventions used in Section 5.

Return Measures

For each return segment n associated with stock j, we compute the Fama–French 25 size-BM characteristic-adjusted return (Fama and French, 1993) over the 20-day *evaluation window* immediately following the *formation window*:

$$R_n^{FF25} = \prod_{s=1}^{20} \left(1 + r_{j,t+s}\right) - \prod_{s=1}^{20} \left(1 + r_{p(j),t+s}\right), \tag{8}$$

where t is the last day of the *formation window* and $r_{p(j),t+s}$ is the daily value-weighted return of the FF25 benchmark portfolio to which stock j is assigned. We evaluate predictability by examining the future FF25-adjusted returns of each segment's $K = 100$ nearest neighbors. The T-statistic is:

$$T_n = \frac{\bar{R}_n^{FF25}}{\hat{\sigma}_n^{FF25}/\sqrt{K}}, \tag{9}$$

where $\bar{R}_n^{FF25}$ and $\hat{\sigma}_n^{FF25}$ are the sample mean and standard deviation of the neighbors' future FF25-adjusted returns. A large $|T_n|$ indicates that segments similar to n have been followed by directionally consistent abnormal returns. These FF25-adjusted returns serve as the primary outcome variable: they are used to compute the T-statistics above and as the prediction target for the engine of Section 5.

An important caveat governs the interpretation of T_n throughout the paper. The neighbors' future returns are not independent observations: neighbors share window end dates—in which case their 20-day evaluation windows overlap entirely—load on common shocks beyond the FF25 adjustment, and occasionally include the same stock at nearby dates. The $\sqrt{K}$ scaling therefore overstates precision, and $|T_n| \geq 1.96$ carries no nominal five-percent size. Accordingly, we treat T_n as a signal-to-noise *score* that ranks segments, and use 1.96 only as a fixed, conventional cutoff chosen ex ante; no result in the paper depends on its calibration. A month-clustered version of the score—averaging neighbor outcomes within each evaluation month, then computing the t-ratio across the roughly ten months represented among a segment's neighbors—quantifies the dependence adjustment: the share of segments crossing 1.96 falls from 10.4% to 7.7%, while the substantive conditioning results are unchanged, as discussed in Section 4.

The Block-Bootstrap Surrogate Benchmark

The surrogate benchmark used in Table A2 replaces the cross-stock shuffle with a null that keeps each stock's own dynamics and the market's factor structure and randomizes only the idiosyncratic shape. It is constructed in four steps.

Factor exposures. For each stock i and each calendar year y we regress the stock's daily excess return on the six Fama–French factors (FF5 plus UMD) and a constant, using that year's daily observations: $r_{i,t} - r_{f,t} = a_{i,y} + \hat{b}'_{i,y} F_t + e_{i,t}$. Exposures are estimated year by year rather than over the full sample, so that a stock's factor loadings are allowed to drift and each year's comovement is absorbed locally. We require more than 200 trading days in the stock-year; stock-years below that threshold have no estimated beta and therefore contribute no surrogate values. The cost of this falls on the benchmark pool rather than on the segments being measured, as set out below.

Idiosyncratic series. The idiosyncratic return is $e_{i,t} = r_{i,t} - \hat{b}'_{i,y} F_t$, the raw return net of the fitted factor component only. The intercept and the risk-free rate are left in, since every 60-day segment is z-normalized before any distance is computed and a constant level is removed there.

Resampling. Each stock's $e_{i,t}$ is resampled independently by the stationary bootstrap of Politis and Romano (1994): block lengths are geometric with success probability $p = 1/L$, so the mean block length is L trading days, and resampling is circular. Blocks are drawn within each maximal run of consecutive trading days for which the stock has a valid beta-year, a run that typically spans several years, and the resampled values are written back to the same calendar dates. Independent scrambling across stocks is innocuous here because the common component is carried entirely by the factor term, which is not resampled.

Reassembly. The surrogate return is $\tilde{r}_{i,t} = \hat{b}'_{i,y} F_t + \tilde{e}_{i,t}$: the factor component is added back, and the realized factor series F_t is kept in its original time order and attached to its own date. Every surrogate stock therefore carries the same real factor path as the data, so the level and timing of comovement in the benchmark panel are identical to the real panel by construction; only the idiosyncratic shape sequence is randomized. We generate three independent replicates, which take the place of the three random-walk benchmark panels. The count is inherited from the baseline design rather than chosen here: the shuffle benchmark averages over exactly three random-walk panels, and a denominator averaged over more draws would carry less sampling noise for a reason having nothing to do with the null, which is precisely the comparison Panel A is built to make. The ratio therefore retains its published form $PDR_n = 3\,\bar{d}_n^{\text{real}\to\text{real}} / \sum_{k=1}^{3} \bar{d}_n^{\text{real}\to\text{surr},k}$, each denominator term being the mean distance to the $K = 100$ nearest neighbors within replicate k. Each replicate supplies one surrogate path for every stock-day that has an estimated beta, so the benchmark pool is the same panel of stocks and dates as the real cross-section, less the stock-years the 200-day rule removes: it holds 11.0% fewer complete 60-day segments (1.15 against 1.30 million). The real segments are unaffected—each is still a query against the pool—and 99.6% of them enter the surrogate analyses.

One difference in granularity is worth making explicit. The bootstrap is run once over the whole panel for each replicate, not year by year, and its blocks are drawn along a stock's own history; the shuffle benchmark it replaces instead permutes returns across stocks within a single year. The two nulls are aligned where the comparison happens rather than where the randomization happens: the surrogate panel is cut into the same annual blocks $\mathcal{D}_y$, carries the same segment identifiers as the real panel, and is run through the same distance and neighbor code, so the PDR values are directly comparable. The main specification sets $L = 10$ trading days; $L = 20$ is reported alongside it, and $L = 5$ was also generated.

Step 1: Landmark Selection (the Recurrence Grid)

For each test year Y, the landmark grid is drawn from the pool of segments in the trailing three years ($Y - 3$ through $Y - 1$), using point-in-time statistics measured against those years' own comparison blocks:

1. **Significance filter ($|T_n| \geq 1.96$).** We retain segments whose neighbor T-statistic (Equation 9) satisfies $|T_n| \geq 1.96$, so that measurement points are placed only where the local region of trajectory space has exhibited a high neighbor-outcome score.

2. **Recurrence stratification (410 per PDR decile).** Among the retained segments we split the pool at its own PDR deciles and draw 410 segments at random from each, so that the grid samples the recurrent and the novel regions of trajectory space evenly rather than concentrating on either. The draw gives up to 4,100 landmarks per test year: 4,100 in forty-nine of the fifty-one test years, and 1,612 and 3,239 in the two earliest, where a decile holds fewer than 410 segments. A year with fewer than 1,000 qualifying segments would be skipped; none is.

Because the draw is random within deciles, five independent grids are drawn per test year and the five resulting predictions are averaged by within-month rank, so that no single draw determines the signal. We denote a resulting landmark set $\mathcal{S}_Y$ and refer to it as the *recurrence grid*. As Sections 5.1 and 5.2 discuss, the estimator's performance is nearly invariant to replacing this grid with random landmarks, consistent with its role as an interchangeable Nyström basis.

Portfolio Formation

Within each month-end date in the test year, we sort all segment observations by the model's predicted score and assign them to five quintile portfolios, Q1 (lowest) through Q5 (highest). We construct two types of portfolios:

- **Equal-weighted (EW):** The portfolio return for quintile q on month-end date τ is the simple average of the holding-period returns of all segments assigned to quintile q:

$$R_{q,\tau}^{EW} = \frac{1}{|Q_{q,\tau}|} \sum_{n \in Q_{q,\tau}} R_n^{raw}, \tag{10}$$

  where $Q_{q,\tau}$ denotes the set of segments in quintile q on date τ, and $R_n^{raw} = \prod_{s=1}^{20}(1 + r_{j,t+s}) - 1$ is the unadjusted cumulative holding-period return. Note that we use *raw* (unadjusted) returns for portfolio evaluation because these represent the actual returns an investor would earn; the FF25-adjusted returns are used only as the prediction target, to ensure that the model targets abnormal performance rather than systematic factor exposures.

- **Value-weighted (VW):** The portfolio return for quintile q on date τ is the weighted average of holding-period returns, where the weights are market capitalizations:

$$R_{q,\tau}^{VW} = \frac{\sum_{n \in Q_{q,\tau}} w_n \cdot R_n^{raw}}{\sum_{n \in Q_{q,\tau}} w_n}, \tag{11}$$

  where w_n is the market equity of segment n's stock at the beginning of the *evaluation window*. To mitigate the influence of extreme weights, we winsorize market capitalizations at the 1st and 99th percentiles within each date before computing the weighted average.

The **long-short portfolio** is constructed as Q5 minus Q1: going long the highest-scored quintile and short the lowest.

Factor Model Specifications

We evaluate the long-short portfolio's risk-adjusted performance by regressing its monthly returns on standard factor models. Let $R_{LS,\tau}$ denote the long-short portfolio return for month τ. We estimate alphas from the following seven models:

1. **CAPM:** $R_{LS,\tau} = \alpha + \beta_{mkt} \cdot MktRf_\tau + \varepsilon_\tau$
2. **Fama-French 3-factor (FF3):** $R_{LS,\tau} = \alpha + \beta_{mkt} \cdot MktRf_\tau + \beta_{smb} \cdot SMB_\tau + \beta_{hml} \cdot HML_\tau + \varepsilon_\tau$
3. **Carhart 4-factor (FF4):** FF3 + $\beta_{umd} \cdot UMD_\tau$
4. **Fama-French 5-factor (FF5):** FF3 + $\beta_{rmw} \cdot RMW_\tau + \beta_{cma} \cdot CMA_\tau$
5. **6-factor (FF6):** FF5 + $\beta_{umd} \cdot UMD_\tau$
6. **FF6 + ST-Rev:** FF6 + $\beta_{strev} \cdot STRev_\tau$, where $STRev$ is French's monthly short-term reversal factor
7. **FF6 + ST-Rev + WST-Rev:** FF6 + ST-Rev + $\beta_{wstrev} \cdot WSTRev_\tau$, where $WSTRev$ is the weekly reversal factor built inside our universe under the strategy's own portfolio rule (Section 5.2)

where $MktRf$ is the market excess return, SMB is the size factor, HML is the value factor, UMD is the momentum factor, RMW is the profitability factor, and CMA is the investment factor. All factor returns except WST-Rev are obtained from Kenneth French's data library; WST-Rev is constructed as described in item 7 above. The intercept α in each regression represents the abnormal return that cannot be explained by the corresponding set of risk factors. All returns and alphas are annualized by multiplying monthly values by 12, and Sharpe ratios are annualized by multiplying the monthly Sharpe ratio by $\sqrt{12}$.

C. Robustness

This appendix collects the robustness evidence for Stage I; the Stage II robustness checks are reported in Sections 5.2 and 5.3. Tables A2 through A4 concern the recurrence measure itself. Table A2 replaces the two choices the recurrence measure rests on—the random-walk benchmark and the nearest-neighbor rule—with a stationary block bootstrap of factor-model residuals and with a diversified neighbor search, and re-runs the downstream analyses against the harder null. Table A3 asks whether PDR persistence is a restatement of volatility persistence. Table A4 varies the length of the pattern window. Table A5 reports the construction of the twelve international samples and the persistence measured in each, and Table A6 varies the pattern window and the holding period together in the sign-preservation test.

Table A2: Robustness: PDR Persistence and Downstream Analyses under Alternative Benchmark and Neighbor Definitions

This table reports the robustness of the one-year-ahead Pattern Distance Ratio (PDR) and of the analyses built on it to two alternative constructions. *Strict Neighbors* replaces the baseline nearest-neighbor search with a diversified search: stocks are partitioned into size × book-to-market × 48-industry × month cells, and at most one neighbor is retained per cell before selecting the $K = 100$ nearest, preventing the neighbor set from being dominated by contemporaneous, same-industry, same-size, same-value peers. *Block-Bootstrap Surrogate* replaces the cross-stock-shuffle benchmark with a stationary block bootstrap of each stock's six-factor (FF5+UMD) residuals; this preserves the stock's real factor exposures—and hence its systematic comovement and the bulk of its own autocorrelation—while scrambling only the idiosyncratic component, yielding a substantially harder null than the cross-stock shuffle; Appendix B gives the full construction. Two mean block lengths are reported: $L = 10$ trading days (the specification discussed in the text) and $L = 20$, under which a 60-day window is stitched from roughly three blocks so that within-window dependence at up to monthly scales—including slow volatility dynamics—is preserved in the null. For each construction we report, across annual year-pairs, the mean Pearson and Spearman correlation between in-sample and out-of-sample PDR; the corresponding cross-year and Newey–West (3 lags) t-statistics and the fraction of year-pairs with a positive correlation are given in the text (Section 3.3). The Baseline and Random Walk columns repeat the main specification for reference. All specifications use the 60-day pattern window at the 1-year horizon, 1971–2025.
Panel B re-runs the downstream PDR analysis at the 1-year horizon with the PDR measured against the $L = 10$ block-bootstrap surrogate instead of the cross-stock shuffle. Because the surrogate preserves each stock's own return distribution, its short-range dependence, and its factor comovement, results that survive it cannot be attributed to distributional heterogeneity, volatility clustering, or comovement. Panel B reports the quintile transition matrix of the $L = 10$ surrogate-based PDR (the shuffle-benchmark counterpart is Panel A of Table 2; under the $L = 20$ surrogate the diagonal is Q1 0.374 / Q5 0.292). Sign preservation under the surrogate is reported in Section 3.3. All entries use 1971–2025.

Panel A: PDR persistence under alternative constructions

Construction	Mean Pearson r	Mean Spearman r
Baseline	0.317	0.233
Strict neighbors	0.350	0.250
Block-bootstrap surrogate ($L = 10$)	0.247	0.176
Block-bootstrap surrogate ($L = 20$)	0.262	0.184
Random walk placebo	0.014	0.010

Panel B: 1-year transition matrix, surrogate-based PDR ($L = 10$)

	Q1	Q2	Q3	Q4	Q5
Q1	0.362	0.195	0.160	0.143	0.140
Q2	0.194	0.226	0.217	0.200	0.163
Q3	0.158	0.215	0.224	0.219	0.185
Q4	0.143	0.200	0.216	0.225	0.216
Q5	0.143	0.165	0.184	0.213	0.296

Table A3: PDR Persistence and the Volatility Channel

This table asks whether PDR persistence is a restatement of volatility persistence. The dependent variable is the one-year-ahead out-of-sample PDR, $PDR_n^{\text{out},1}$; the regressors are the segment's in-sample PDR and four controls measured on the same 60-day formation window: idiosyncratic volatility, the standard deviation in daily units of the residual from regressing the stock's daily return on the Fama–French market return over those same 60 days; log market equity on the last day of the window, the compound return over the last 21 days, and the slope of the cumulative log-price path. Each year-pair cross-section is estimated separately with every variable standardized within that cross-section; the reported coefficients are Fama–MacBeth averages over the 54 year-pairs, with Newey–West (3 lags) t-statistics in parentheses. Avg. R^2 is the mean of the per-cross-section R^2 rather than a pooled R^2, and the three columns are estimated on one common sample, so the observation count is the same in each. The lower panel reports the mean year-pair persistence correlation within terciles of that same in-sample idiosyncratic volatility, formed separately within each year-pair so that each tercile holds a third of that year-pair's segments. ***, **, * denote significance at the 1%, 5%, and 10% levels.

	Dependent variable: $PDR_n^{\text{out},1}$		
Regressor	(1)	(2)	(3)
In-sample PDR (PDR_n^{in})	0.317*** (8.08)	0.304*** (7.87)	0.304*** (7.85)
IVOL		0.093*** (9.11)	0.095*** (7.28)
Log ME			0.003 (0.26)
Reversal (monthly, 21-day)			0.017*** (3.26)
Trend			0.002 (0.20)
Avg. R^2	0.125	0.136	0.145
Observations	1,295,590	1,295,590	1,295,590
Mean persistence correlation within IVOL terciles			
Low IVOL	0.215 (positive in 49 of 54 year-pairs)		
Mid IVOL	0.270 (positive in 53 of 54 year-pairs)		
High IVOL	0.389 (positive in 54 of 54 year-pairs)		

Table A4: Robustness: PDR Persistence by Pattern Window Length

This table reports the robustness of Pattern Distance Ratio (PDR) persistence across the full grid of pattern windows $\{30, 60, 90\}$ days crossed with holding periods $\{20, 30\}$ days. The baseline is the 60-day window with a 20-day holding period. The two dimensions are not symmetric. The PDR is a ratio of nearest-neighbor distances between standardized trajectories, so the holding period does not enter its construction at all; it can affect these statistics only by changing which segments qualify, because a segment must have a complete window of window + holding days. That channel is negligible: within a given window length the Pearson correlation moves by at most 0.004 across holding periods, against a 0.046 spread across window lengths. Panel A presents overall correlation statistics between in-sample and out-of-sample PDR (N in thousands of segment pairs). Panel B reports year-by-year consistency metrics. The holding period matters materially only for the neighbor-outcome T-statistic, whose sensitivity is reported in Table A6. All specifications use 1-year rolling windows with data from 1971–2025.

Panel A: Overall Correlation—Pooled

	30-day pattern		60-day pattern		90-day pattern	
Metric	20d	30d	20d	30d	20d	30d
Pearson r	0.333	0.330	0.308	0.304	0.287	0.287
Spearman r	0.289	0.288	0.251	0.250	0.240	0.240
N (000s)	1,314	1,302	1,296	1,283	1,250	1,250

Panel B: Year-by-Year Consistency

	30-day pattern		60-day pattern		90-day pattern	
Metric	20d	30d	20d	30d	20d	30d
Mean Spearman	0.274	0.273	0.233	0.232	0.209	0.209
Std Spearman	0.157	0.157	0.124	0.123	0.118	0.118

Table A5: International Samples and PDR Persistence Across Markets

This table summarizes the construction of the international samples used in the PDR persistence robustness (Section 3.3) and in the out-of-sample engine deployment (Section 5.6, Table 8), and reports PDR persistence in each market.

Sample construction. Daily security data are from Compustat Global (`g_secd`), full available history per market. Screens: common shares only (issue type code 0); one primary issue per firm; daily total returns constructed from adjusted closing prices and total-return factors; month-end market equity converted to U.S. dollars at daily exchange rates and required to exceed US$10 million. Return segments are 60 trading-day windows anchored at month-end dates and require a complete window. *Segments* is the total number of 60-day segments; *Stocks/month* is the average number of scored stocks per month in the engine's out-of-sample period; *Engine OOS* is the out-of-sample prediction window, which begins when the expanding training sample first reaches the minimum size and ends in December 2025.

PDR persistence. The PDR is constructed exactly as in the U.S. analysis, three-shuffle random-walk benchmark included (Section 2.3), and persistence is measured for each year y as the cross-sectional correlation between PDR^{in} (from $\mathcal{D}_y$ vs. $\mathcal{D}_y$) and the one-year-ahead PDR^{out} (from $\mathcal{D}_y$ vs. $\mathcal{D}_{y+1}$). *Mean* r and t are the time-series mean and cross-year t-statistic of the annual Pearson correlations; *% Pos* is the fraction of country-years with a positive correlation; *Pooled* r and *Pooled* ρ are computed over the pooled cross-section of all retained country-years. We retain country-years with at least 5,000 overlapping pattern observations (N is their pooled total), which omits seven markets that lack the cross-sectional depth (Brazil, Switzerland, Ireland, Israel, Italy, Poland, and Singapore), and we report only markets in which both stages of the paper can be evaluated, which additionally omits the United Kingdom (persistence 0.34, $t = 4.2$, but its Stage II long–short series is populated in only 71 of the 240 months it spans). ***, **, * denote significance at the 1%, 5%, and 10% levels.

	Sample construction			PDR persistence						
Market	Segments	Stocks/ month	Engine OOS	Years	N	Mean r	t	% Pos	Pooled r	Pooled ρ
Hong Kong	184,219	825	2010–2025	15	153,397	0.471***	11.10	100.0	0.507	0.452
Thailand	109,861	539	2013–2025	11	74,832	0.429***	11.53	100.0	0.430	0.372
Malaysia	116,627	356	2004–2025	7	40,208	0.392***	5.33	100.0	0.379	0.344
Japan	898,892	2,405	1995–2025	34	879,743	0.381***	13.37	100.0	0.465	0.337
France	144,506	429	2003–2025	14	77,711	0.310***	9.15	100.0	0.309	0.239
South Korea	477,557	1,612	2003–2025	25	452,549	0.305***	9.90	100.0	0.420	0.327
Australia	126,531	407	2005–2025	8	49,824	0.295***	7.00	100.0	0.283	0.201
Germany	89,734	332	2009–2025	6	32,058	0.258***	5.28	100.0	0.258	0.222
India	282,776	1,545	2012–2025	14	256,379	0.202***	8.15	100.0	0.211	0.198
Sweden	83,145	314	2009–2025	6	35,390	0.196***	3.99	83.3	0.180	0.156
Taiwan	358,273	1,260	2004–2025	23	333,512	0.136***	5.41	82.6	0.129	0.132
China	586,576	2,251	2006–2025	20	544,758	0.089***	3.12	75.0	0.143	0.129
Total	3,458,697	—	—	—	—	—	—	—	—	—

Table A6: Robustness: Sign Preservation Rate by Pattern Length and Holding Period

This table reports the robustness of sign preservation rates to alternative pattern lengths and holding periods. Panel A presents results for segments with high neighbor-outcome scores ($|T| \geq 1.96$), and Panel B for segments with low scores ($|T| < 1.96$). We report the full grid of pattern windows $\{30, 60, 90\}$ days crossed with holding periods $\{20, 30\}$ days. The baseline specification is the 60-day pattern with a 20-day holding period. Entries are in percent. Sign preservation measures the proportion of segments for which the sign of the T-statistic is the same in-sample and out-of-sample. The "Adj. Null" row reports the independence-adjusted null benchmark, computed as $P(\text{in}>0) \times P(\text{out}>0) + P(\text{in}<0) \times P(\text{out}<0)$, where $P(\text{out}>0)$ is the unconditional probability from the full sample. All specifications use 1-year rolling windows with data from 1971–2025.

Panel A: High Statistical Significance ($|T| \geq 1.96$)

	30-day pattern		60-day pattern		90-day pattern	
Quintile	20d	30d	20d	30d	20d	30d
Q1 (Low PDR)	53.40	52.09	53.07	52.44	52.22	51.12
Q2	53.42	51.96	53.56	53.00	52.19	51.30
Q3	52.98	51.74	53.08	52.42	52.06	51.49
Q4	52.85	51.83	52.84	51.87	51.57	50.93
Q5 (High PDR)	49.56	49.66	50.63	49.72	50.84	50.66
Adj. Null	50.22	50.18	50.24	50.17	50.11	50.26

Panel B: Low Statistical Significance ($|T| < 1.96$)

	30-day pattern		60-day pattern		90-day pattern	
Quintile	20d	30d	20d	30d	20d	30d
Q1 (Low PDR)	50.97	50.66	51.13	50.83	50.67	50.58
Q2	51.43	50.73	51.11	50.54	50.83	50.61
Q3	51.13	50.90	51.04	50.75	50.83	50.52
Q4	50.75	50.55	50.92	50.68	50.62	50.30
Q5 (High PDR)	50.38	50.21	50.46	50.31	50.48	50.21
Adj. Null	50.02	50.01	50.02	50.00	50.00	50.02

Table A7: The Horse Race, Value-Weighted

This table repeats Table 6 with value-weighted Fama–MacBeth regressions: each month's cross-sectional regression is estimated by weighted least squares with market equity as the weight (Medhat and Schmeling, 2022; Green et al., 2017). The columns, the sample, the variable definitions, the winsorization and standardization, and the Newey–West (12 lags) t-statistics are exactly as in Table 6; only the weighting differs. A value-weighted regression on the signal alone gives a slope of 0.175 (t = 3.91). ***, **, * denote significance at the 1%, 5%, and 10% levels.

Regressor	(1)	(2)	(3)	(4)
Signal (engine prediction)	0.139*** (3.54)	0.076*** (3.11)	0.089*** (3.55)	0.072*** (3.00)
MA ratio	−0.999*** (−5.29)	−0.974*** (−6.48)	−0.513*** (−3.29)	−0.495*** (−3.19)
MA crossover	0.936*** (4.94)	0.813*** (5.78)	0.376*** (2.70)	0.342** (2.32)
Reversal (monthly, 21-day)		−0.082 (−1.43)	−0.001 (−0.02)	0.013 (0.22)
Momentum (12–2)		0.373*** (4.83)	0.360*** (4.75)	0.387*** (5.40)
Long-term reversal (60–25)		−0.077* (−1.73)	−0.077* (−1.76)	−0.076* (−1.77)
IVOL		−0.284*** (−3.62)	−0.276*** (−3.62)	−0.289*** (−4.04)
Max		0.062 (1.43)	0.063 (1.47)	0.065 (1.61)
Beta		0.069 (0.79)	0.084 (1.00)	0.043 (0.51)
log market equity		0.068 (0.46)	0.058 (0.39)	0.030 (0.21)
log book-to-market		0.007 (0.15)	0.003 (0.07)	−0.009 (−0.21)
Illiquidity (Amihud)		0.162** (2.57)	0.160** (2.57)	0.161*** (2.59)
Turnover		0.123* (1.65)	0.122* (1.67)	0.123 (1.62)
Bid–ask spread		0.010 (0.16)	−0.013 (−0.20)	0.008 (0.15)
Dollar volume (log)		−0.119 (−0.88)	−0.113 (−0.85)	−0.088 (−0.65)
Zero-volume days		−0.045 (−1.47)	−0.044 (−1.49)	−0.047 (−1.60)
Trend forecast (HZZ)				0.012 (0.18)
Price delay				−0.004 (−0.13)
52-week high		−0.192** (−2.55)	−0.212*** (−2.82)	−0.220*** (−2.99)
Reversal (weekly, 5-day)			−0.216*** (−4.66)	−0.213*** (−4.23)
Avg. adj. R^2 (%)	5.57	20.20	20.65	21.41

D. A Latent-Template Model of Pattern Recurrence

This appendix formalizes the density-ratio interpretation of the PDR (Section 2.4) in a simple statistical model of recurring return trajectories and derives the empirical implications tested in Sections 3–5. The model is deliberately reduced-form: "templates" stand in for any economic mechanism that causes particular return dynamics to repeat across stocks and over time—earnings and announcement cycles, institutional rebalancing, the gradual diffusion of information, or the crowding of trading strategies. The framework's role is to make precise *what* the PDR identifies, not to explain *why* recurrence arises.

Setup

After z-normalization, each m-day return segment ($m = 60$) is a point in the shape space

$$\mathcal{X} \;=\; \left\{x \in \mathbb{R}^m : \textstyle\sum_{i=1}^m x_i = 0,\; \|x\|^2 = m - 1\right\},$$

a sphere of dimension $p = m - 2 = 58$. We model the segments in a comparison block as independent draws from a two-part mixture.

Assumption A1 (latent-template mixture). With probability $1 - \pi$, a segment is a draw from a diffuse null density f_0 on $\mathcal{X}$—the shape distribution induced by temporally unstructured, cross-sectionally pooled returns. With probability π, it is a noisy realization of one of M latent templates: template k is selected with weight w_k and generates the segment from a density f_k concentrated near a location $\tau_k \in \mathcal{X}$. The density of real segments is therefore

$$f(x) \;=\; (1 - \pi)\, f_0(x) \;+\; \pi \sum_{k=1}^{M} w_k\, f_k(x), \tag{12}$$

and we define the *local template intensity*

$$\lambda(x) \;=\; \frac{\sum_k w_k\, f_k(x)}{f_0(x)}, \tag{13}$$

the template mass near x measured relative to the generic commonness of the shape x under the null.

Assumption A2 (benchmark validity). The random-walk segments are draws from f_0, in a comparison set of the same size N as the real block. This holds by construction: the benchmark reshuffles the same pool of returns within the same calendar year, applies the same within-window standardization, and is searched with the same K nearest neighbors (Section 2.3).

Assumption A3 (regularity). f_0 and f are continuous and positive on $\mathcal{X}$, and the nearest-neighbor asymptotics $K \to \infty$, $K/N \to 0$ apply as $N \to \infty$.

Lemma 1 (nearest-neighbor distances estimate local density). *Under A3, for a query point x and a comparison sample of N draws from a density g, the mean distance to the K nearest neighbors satisfies*

$$\bar{d}_K(x) \;=\; c_p \left(\frac{K}{N\, g(x)}\right)^{1/p} \big(1 + o_p(1)\big), \tag{14}$$

where c_p is a dimensional constant. Nearest-neighbor distances are thus inversely related to the local density $g(x)$ (Loftsgaarden and Quesenberry, 1965).

What the PDR identifies

Proposition 1 (PDR is a local density ratio; behavior under the null). *Under A1–A3, applying Lemma 1 to the numerator of the PDR (comparison density f, sample size N) and to its denominator (comparison density f_0, sample size N),*

$$\operatorname{plim} PDR(x) \;=\; \left(\frac{f_0(x)}{f(x)}\right)^{1/p} \;=\; \Big[(1-\pi) + \pi\,\lambda(x)\Big]^{-1/p}. \tag{15}$$

The constants c_p, K, and N cancel exactly. In particular, if $\pi = 0$ (no recurring structure), then $f = f_0$ and $PDR(x) \to 1$ for every x; since the population PDR is then constant, sample PDRs computed on disjoint comparison blocks are independent estimation noise, and the correlation between in-sample and out-of-sample PDR converges to zero.

The empirical counterpart of the null case is the random walk placebo, whose pooled in-sample/out-of-sample correlation is 0.011 (Table 1) against 0.308 in real data, and whose transition matrices have all entries near 0.20 (Table 2, Panel D). The proposition predicts a population correlation of exactly zero; a finite placebo instead delivers a small realization-specific residual—across the twenty configurations that can be formed from our five independent shuffled series the mean correlation ranges only from −0.003 to +0.017, while the corresponding conventional t-statistics range from −0.5 to +3.5—so the informative comparison is one of order of magnitude, not of statistical significance.

Remark 1 (why the raw neighbor distance is not a recurrence measure). By Lemma 1, the numerator alone satisfies $\bar{d}_n^{real\to real} \propto (K/(Nf(x)))^{1/p}$: it depends on the total local density f, and hence on the generic commonness $f_0(x)$ of the segment's shape and on the comparison-set size N. Sorting on the numerator therefore sorts jointly on shape commonness, sample density, and recurrence. The ratio in Equation (15) is free of $f_0(x)$, N, K, and c_p, and depends on the data-generating process only through $\lambda(x)$—template mass *relative to* generic commonness. This is the formal content of the discussion in Section 2.4.

Proposition 2 (two-sided detection and compression). *Under A1–A3: (i)* $\operatorname{plim} PDR(x) < 1$ *if and only if $\lambda(x) > 1$, and the limit is strictly decreasing in $\lambda(x)$: low PDR identifies regions of trajectory space where real data are denser than chance implies. (ii) In template-free regions ($\lambda(x) \approx 0$),* $\operatorname{plim} PDR(x) = (1-\pi)^{-1/p} > 1$*: because a fraction π of real mass is drawn into template regions, segments far from every template are* less *matched in real data than in the benchmark. High PDR is therefore informative uniqueness, not noise. (iii) Deviations from one are compressed by the exponent $1/p$: for small $\pi(\lambda-1)$,* $\operatorname{plim} PDR(x) \approx 1 - \pi(\lambda(x)-1)/p$.

Part (iii) explains why the empirical PDR distribution is tightly concentrated around one—mean PDR ranges only from 0.982 in the lowest quintile to 1.038 in the highest (Table 1, Panel A)—while its *ranking* remains highly informative: with $p = 58$, even substantial template intensity moves the level of the ratio only slightly, so small stable differences in PDR reflect large differences in local density.

Persistence

Proposition 3 (persistence of recurrence). *Suppose the template structure (τ_k, w_k, π) is stable across comparison blocks. Then the population PDR, $\psi(X_n) = [(1-\pi) + \pi\lambda(X_n)]^{-1/p}$, is a fixed attribute of a segment, and estimated PDRs on disjoint blocks satisfy $\widehat{PDR}_n^{in} =$*

$\psi(X_n) + e_n^{in}$ and $\widehat{PDR}_n^{out} = \psi(X_n) + e_n^{out}$ with independent estimation errors. Consequently: (i) $\operatorname{corr}(\widehat{PDR}^{in}, \widehat{PDR}^{out}) = \operatorname{Var}(\psi)/[\operatorname{Var}(\psi) + \operatorname{Var}(e)] > 0$*; (ii) the correlation rises as estimation noise falls, i.e., with the effective size of the comparison sample; (iii) under the null,* $\operatorname{Var}(\psi) = 0$ *and the correlation is zero; (iv) quintile retention is highest where ψ is most distinct from its median relative to noise—the two tails of the λ distribution.*

Part (i) is the persistence documented in Table 1; part (ii) is confirmed directly by the block-length specifications: lengthening the comparison blocks from one to three years enlarges the comparison sample and raises the correlation from 0.308 to 0.486, while holding the blocks fixed at one year the correlation declines only gently with temporal distance (0.308 to 0.270; Table 1, Panel D); part (iii) is the placebo. Part (iv) matches the U-shaped diagonal of the transition matrices (Table 2): both extremes persist more than the middle quintiles, and the template-free region is the most stable of all—$\lambda(x) = 0$ exactly, with no template weights to drift—consistent with Q5 retention (0.41–0.49) exceeding even Q1 retention (0.27–0.31).

Predictability

Assumption A4 (templates carry return information). Template k is associated with a conditional mean μ_k of the future characteristic-adjusted return of segments it generates; null draws have conditional mean zero.

Proposition 4 (recurrence and predictability). *Under A1–A4, the expected future adjusted return of a segment at location x is*

$$E\big[r \mid x\big] \;=\; \sum_{k=1}^{M} P(k \mid x)\,\mu_k, \qquad P(k \mid x) = \frac{\pi w_k f_k(x)}{f(x)}, \tag{16}$$

and the total posterior template probability $q(x) = \pi\lambda(x)/[(1-\pi) + \pi\lambda(x)]$ is monotone decreasing in the population PDR. Consequently: (i) averages of nearest neighbors' realized outcomes (the neighbor T-statistic of Appendix B) estimate the local conditional mean and are predictive out of sample; (ii) the reliability of that prediction increases with $q(x)$ and therefore decreases with PDR: recurrence strength and predictability are linked; (iii) $E[r \mid x]$ is a smooth function of location in shape space, so it can be approximated by ridge regression on kernel similarities to a fixed set of landmarks whose coverage includes the template regions—the Section 5 estimator; permuting the training targets breaks the correspondence between location and μ_k and drives predictability to zero; (iv) under the null, $E[r \mid x] \equiv 0$: there is no pattern-based predictability to find.

Part (ii) corresponds to the declining gradient in sign preservation across PDR quintiles (Table 3) and to the rank-IC gradient across PDR terciles for the trading signal (Section 5.3); part (iii) corresponds to the scrambled-target placebo (Section 5.3).

Remark 2 (the landmark basis is interchangeable). Proposition 4(iii) requires of the landmarks only that their coverage include the template regions; it does not require that they be *selected* on recurrence. The information the estimator exploits resides in the training pairs of trajectories and realized outcomes—the landmarks merely fix the finite-dimensional basis in which the kernel regression is expressed—so any landmark set that covers the support of the data spans nearly the same space of smooth functions of location. Moreover, because real segments are drawn from the mixture f of Equation (12), a landmark set drawn at random from the real data allocates mass to template regions in proportion to their density—the density structure documented in Section 3—so random real-data landmarks are not a recurrence-free alternative to the recurrence grid. The near-identical alphas under the two bases (Section 5.2) are therefore the predicted behavior of a basis choice, not evidence that recurrence is unrelated to the predictability; the recurrence–predictability link is tested instead by the landmark-independent implications of Proposition 4(ii)—the gradients of Table 3 and of Section 3.3.

Discussion

Remark 3 (misspecification of the null). The cross-stock shuffle removes each stock's own time-series dependence *and* homogenizes the cross-section, so if returns exhibit univariate autocorrelation (e.g., volatility clustering) or persistent stock-level distributional heterogeneity (stocks with systematically different tail, skewness, or jump behavior), the empirical benchmark understates f_0 in the affected regions and the PDR can flag recurrence that reflects only own-stock dependence or distribution type. The block-bootstrap surrogate of Section 3.3 re-estimates the benchmark under a null that preserves own-stock autocorrelation and factor exposures; persistence survives at roughly three quarters of its baseline strength, indicating that the template structure is not an artifact of the choice of f_0.

Remark 4 (economic content). The model is silent on the origin of templates, which is precisely what makes it a framework rather than a theory: any mechanism that generates repeated conditional return dynamics—behavioral, institutional, or informational—maps into some (τ_k, w_k, μ_k). The evidence in Section 5.4 (the concentration of signal quality in small and high-spread stocks, and the post-2000 attenuation of the short-leg correction) suggests that at least part of the template-linked return predictability behaves like mispricing that arbitrage competes away, though, as discussed there, a risk-based account cannot be ruled out.

In sum, the framework maps one-to-one onto the empirical design of the paper: Proposition 1 underlies the random walk placebo; Proposition 2 rationalizes the tight but informative PDR distribution and the informativeness of uniqueness; Proposition 3 delivers the persistence and transition-matrix results of Section 3; and Proposition 4 delivers the recurrence–predictability link of Section 4 and the design and placebo of the portfolio strategy in Section 5.